CTS
Centre for Technology and Society
UNIVERSITY OF OXFORD

WORKING PAPER SERIES
WP 2026-02

Tariffs, Geoeconomics and Political Cleavages

Evidence from American High-Tech Firms

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AUGUST 2026

ABSTRACT

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Tariffs · Geoeconomics · American high-tech companies

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Abstract

Standard trade theories predict that capital-intensive firms deeply integrated into global production networks would strongly oppose the Trump administration's announcement of sweeping global tariffs on the "Liberation Day" of April 2nd 2025. Why, then, do American high-technology firms respond with such heterogeneity? We argue that firms that play central roles in US geoeconomic influence strategies receive special treatment, in return for more moderate position toward tariff policy. We test this argument using a new firm-quarter panel of 486 earnings-call transcripts from 65 leading U.S. technology and defense firms (2024Q1–2026Q1), with firm-level measures of tariff negativity and upstreamness generated through a two-step LLM-based coding pipeline (GPT-4o). Incorporating these measures into a difference-in-differences design that leverages Liberation Day as an exogenous shock, we find strong evidence for our theory: hardware firms respond more negatively than digital firms, and more upstream hardware firms respond less negatively than their downstream counterparts. Lobbying disclosure data corroborate these patterns, and suggestive evidence for the special treatment shows that upstream products are more likely to get exempted from 301 tariffs during 2017-2020.

Keywords: Tariffs; Geoeconomics; American high-tech companies

*We want to thank Jerik Cruz, Karl Grindal, Sanghyun Han, Jenny Jun, In Song Kim, Ranjit Lall, Eddy Malesky, Ron Rogowski, Ryan Weldzius, Chris Whyte and audiences at the Workshop on Disruptive Technology and Geopolitical Upheaval, Institute for Humane Studies for their helpful feedback and insights. We also thank Athena Haq for excellent research assistance.

Introduction

President Donald Trump announced sweeping tariffs on imports from virtually every country in the world in April 2025, which triggered a range of responses from American companies. The American high-tech sector was among the most sensitive to tariffs because of its deep integration into global value chains, but firms' attitudes within the industry were far from homogeneous. Qualcomm, despite a strong supplier relationship with Apple and similar surface-level exposure to the global market, was comparatively unconcerned about the impact of tariffs: in April 2025, Qualcomm said that it did not expect "any material impact from tariffs, and that it had not seen elevated buying of its products ahead of tariffs during the quarter" (Leswing 2025). Apple, by contrast, expressed considerable concern. In May 2025 Tim Cook told analysts that tariffs could increase Apple's supply costs by $900 million in the June quarter (Business 2025). This divergence reflects a broader pattern. High-tech companies share many characteristics: they are capital-intensive and productive, highly exposed to foreign markets, and generators of high-value-added products, yet they react differently to tariffs.

Such divergence is, to some extent, irreconcilable with studies that anchor trade-policy preferences to these firm-level dimensions (Hiscox 2002; Rogowski 1990; Zhang 2025; Osgood 2018). This literature has little to say about within-sector variation in firms' responses to negative shocks. These studies would predict a unified response among high-tech firms, since such firms are capital-intensive, skill-intensive, deeply integrated into global production networks, and highly reliant on the global market, either for importing intermediate inputs or exporting finished products. Why do some firms display ambivalence, or even support, toward tariffs? The leading firm-level theory of trade pref-

erences similarly fails to account for this pattern. According to the Melitz (2003) model, high-productivity firms thrive under free trade, while low-productivity firms survive in protected markets. US high-tech firms are globally competitive and are among the most productive firms in the world. They should staunchly defend free trade.

Recent developments in the study of economic statecraft offer the building blocks of an explanation for the above puzzle. Leading states/hegemons may seek to exert influence by manipulating the behavior of firms, and leading high-tech firms have technology, capital, expertise, and market power, which are integral to power projection (Clayton, Maggiori, and Schreger 2023). How effective the projection is depends on cooperation and information sharing by private actors, so states rely on both inducements and coercion as the way to mobilize firms and extract values from their coordination against an adversary. Since firms vary in their products and positions in global value chains, they play different roles in state strategy. This variation generates a corresponding range of inducements, producing heterogeneous firm responses to any given policy.

Based on the formal work of Clayton et al. (2023), we propose a firm-level framework of relevance to statecraft, which we call *geoeconomic centrality*. This concept combines what Clayton et al. (2023) call *Micro Power*, which is the ability of the state to coordinate the actions of firms, and *Macro Power*, which is the state influence gained thereby over foreign entities, and these two types of power are the theoretical backbone for our firm-level argument. Firms that are geoeconomically central are those that are relatively easy to coordinate, which generally means that they produce high-value, differentiated, tangible goods that are easy to track and control. In addition, they must occupy upstream positions in supply chains so that withholding their products disrupts production

networks.[1] We expect that firms occupying positions of geoeconomic centrality moderate their criticism of government policy during episodes of economic statecraft because they receive some forms of special treatment from the state for the adverse effects of that policy.

We test our argument with a two-step empirical strategy. First, we construct a new firm-quarter panel dataset of 486 earnings-call transcripts and quarterly lobbying disclosure reports from 65 Fortune 500 technology and defense firms, covering nine consecutive quarters from Q1 2024 through Q1 2026. Using a two-stage large language model pipeline, we generate firm-level measures of expressed tariff negativity and the degree of upstreamness from the substantive content of each earnings-call transcript. The LLM-coded approach allows us to capture the nuance of corporate communication on trade policy at a scale beyond what manual coding or dictionary methods can achieve, and we validate the resulting measures through face validity checks and within-firm consistency tests across quarters. Second, we leverage the Liberation Day tariff announcement of April 2, 2025 as an exogenous shock to the trade policy environment to design a difference-in-differences setting to identify the causal effect of geoeconomic centrality on tariff negativity. The results indicate that the hardware firms most directly affected by state influence strategies react most negatively to the tariff shock, but among them, the upstream ones moderate their opposition.

Our contributions are twofold. First and most importantly, we offer an alternative theoretical lens for the cleavage of firms in their perception of tariffs: geoeconomics. The impact of geoeconomic instruments on the behavior of firms is understudied in the

1. The Clayton et al. (2023) model is an explicitly supply-side model of economic influence.

literature of IPE and security. Being at the forefront of great-power geoeconomic competition, firms, especially those in strategic sectors, are central to the state's geoeconomic apparatus (Clayton, Maggiori, and Schreger 2026). Understanding how they respond to geoeconomic instruments is thus essential to the study of the (geo)politics of business, and more broadly to the understanding of economic security. By presenting a geoeconomic explanation for clevages among high-tech firms, we challenge a line of research that predicts homogeneous responses to tariffs, given their similarities in export orientation, productivity and GVC positions. Methodologically, we contribute to the emerging literature using LLMs and text analysis to extract latent business information (Seiler 2025; Ash, Gauthier, and Widmer 2024; Caldara and Iacoviello 2022; Andreou, Harris, and Philip 2020). While we do not pioneer the use of text analysis and earnings-call transcripts for studying corporate strategy or political attitudes, we are, to the best of our knowledge, among the first to link geoeconomics, tariffs, and text analysis. Our contribution thus sets the ground for future work incorporating text data into the study of geoeconomics and business opinion.

We proceed as follows. First, we critically survey relevant existing studies to situate our research in the literature. Based on that, we present our theoretical framework and key testable hypotheses. In the following section, we detail our research design, including our key data and variables, empirical threats, and identification strategy. Next, we discuss our main results and their substantive meaning, which are foundational to the policy implications. Finally, we conclude by summarizing the importance of our results for both policymakers and scholars.

Economic Tension, Tariffs and MNC Behavior

As competition between the United States and China has intensified, both sides have used economic tools to coerce other actors into aligning with their interests or to weaponize their centrality in the global economic network (Farrell and Newman 2019; Gertz and Evers 2020; Cha 2023; Norris 2016; Lee and Maher 2022). These measures include tariffs, trade sanctions, export controls, strategic subsidies, and investment screening (Bauerle Danzman and Meunier 2024; Blackwill and Harris 2016; Mulder 2022; Eaton and Engers 1992; Mastanduno 2020; Hirschman 1980; Clayton, Maggiori, and Schreger 2024). The consequences have included reduced aggregate real income in both the United States and China (Fajgelbaum and Khandelwal 2022), divestment from China by multinational corporations (Vortherms and Zhang 2024), and shifting political coalitions in the United States (Mansfield and Solodoch 2024; Kim and Margalit 2021).

Against this backdrop, firms are indispensable, and states often mobilize their companies to achieve economic coercion goals. Clayton et al. (2023) formalize, in a game-theoretic framework, how states weaponize firms. According to their model, the ability to control domestic firms that provide upstream inputs into global value chains is a key component of hegemonic power. Other studies show, however, that firms do not internalize the foreign-policy interests that motivate states to engage in geoeconomic influence, and instead have incentives to resist policies that raise their input costs or restrict their access to international markets. Chen & Evans (2023) argue that firms support or oppose their home state's use of economic statecraft depending on whether they are high-value or low-value businesses. Other work shows that firm responses to economic sanctions diverge: some over-comply, while others engage in sanctions-busting (Early and Peterson

2024; Pham 2025). Private actors actively influence policymaking processes. Firms originating from states targeted for economic reasons, or sharing strategic traits with existing targets, are more likely to engage in lobbying efforts to influence US sanctions (Peksen and Peterson 2023; Crescenzi 2018). Similarly, prominent chipmakers including NVIDIA, Qualcomm, and Intel have lobbied against export controls on China, leading the Trump administration to relax the regime.[2]

One of the most salient issues for firms is tariffs. When a state imposes tariffs targeting an adversary, its own domestic firms often bear costs, especially in the short to medium term, due to higher input costs, disrupted supply chains, and potential retaliatory measures. Consequently, firms act collectively to shape tariffs in their favor, conditional on their characteristics. Following that logic, recent work in political science has examined more granular variation in firms' lobbying behavior on tariffs. Kim (2017), offering a theory of trade liberalization, finds that productive exporting firms, especially those with differentiated products, are more likely to lobby against tariffs, and that highly differentiated products face lower tariff rates. In the context of the U.S.-China trade war, Eldes et al. (2025) show that lobbying for tariff reductions by large firms is substantially more effective than for small firms, except for firms with foreign subsidiaries in China. These results emphasize the proactive role of firms in crafting tariff policy, and demonstrate some success in shaping tariff outcomes in their favor.

Notwithstanding their valuable contributions, previous research has yet to explain the puzzle we raise: the varying firm-level responses to tariffs under geopolitical tension. Existing studies have not established a direct link between geopolitics, tariffs, and firm be-

2. "Nvidia shares climb as White House relaxes export rules on China chip," *Financial Times*, July 14, 2025. https://www.ft.com/content/ba0929bd-5912-44fb-9048-c143aced4c8a?

havior. Multinational corporations (MNCs), embedded in complex and interdependent production networks, are highly vulnerable to geopolitical disruptions. Consequently, their decision-making abroad reflects not only profit-maximization but also geopolitical calculations. To mitigate geopolitical risks, including those arising from tariffs, firms adopt a variety of strategic responses (Caldara and Iacoviello 2022; Lubinski and Wadhwani 2020; Abdelal 2015; Bednarski et al. 2024). We propose that a firm's geoeconomic centrality helps explain the variation in firm responses to tariffs. Specifically, we highlight two mechanisms determining how geoeconomically central a firm is: its chokepoint control - how costly a firm's absence is to an adversary's economy - and its mobilizability - the likelihood of being mobilized or targeted for states' geopolitical purposes.

Theory & Hypotheses

Scope Conditions & Assumptions

Given the salience of economics in great-power geopolitical competition, firms are indispensable to state power, enabling their states to coerce opponents through market channels (Clayton, Maggiori, and Schreger 2024; Kalyanpur and Newman 2019; Clayton, Maggiori, and Schreger 2023). Accordingly, our theory rests on three scope conditions. First and most importantly, we primarily look at high-tech sectors. Technology is the main arena of contemporary geopolitical rivalry (Ding 2024; Schmidt 2023), and technological innovation is largely driven by firms, especially leading ones (Dosi 1982; Cantwell and Fai 1999; Malerba 2002; Gereffi, Humphrey, and Sturgeon 2005). High-tech firms occupy various positions in fragmented global value chains, from upstream suppliers to

downstream assemblers, and those in central positions can complement state coercive power, making them highly susceptible to geoeconomic mobilization. This dynamic is not unique to high-tech, but it extends to other strategic sectors where firms control means of production, which generates similarly fragmented yet interdependent GVCs. Second, we confine our theory to great-power competition. Great powers usually have influential firms in high-tech industries, and these powers have the ability and will to mobilize their firms to achieve strategic goals (Farrell and Newman 2019). Middle and smaller powers, while sometimes home to powerful companies, often lack the ambition and means to mobilize these firms for geopolitical ends. Narrowing the scope to great-power competition thus ensures that the variation in our outcome variable stems from a firm's position in global value chains and its degree of state mobilization, rather than from confounders such as differences in states' structural capacity to use firms as geopolitical instruments. Additionally, we focus on private, multinational firms (MNCs) rather than firms under direct state control. Private firms must balance profit-maximization, shareholder pressures, and global market access against the political and regulatory constraints imposed by their home states, so their support for tariffs or any such policy instrument cannot be taken for granted. In this way, private firms provide the clearest setting in which we can observe how our independent variables translate into the outcomes of interest.

Our theory relies on three key assumptions. For analytical feasibility, we assume that, over a relatively short-to-medium time horizon, firms are rational and forward-looking. This assumption is plausible given our scope conditions above. High-tech industries are naturally fast-paced and disruptive, incentivizing companies to prioritize immediate, foreseeable payoffs over uncertain long-term outcomes. Moreover, under conditions of

extreme strategic uncertainty, where the duration of tariffs, the scope of retaliation, and the trajectory of great-power rivalry are genuinely unpredictable, firms cannot reliably optimize over very long time horizons. As a result, they are more inclined to weight near-term payoffs and observable policy signals more heavily than speculative long-run returns (Knight 1921; Ellsberg 1961; Beckert 2016). Furthermore, we treat the timing of tariff announcements as plausibly exogenous even if the tariffs themselves are not. While firms may anticipate tariff policy in general, the specific timing of announcements is driven by factors unrelated to firm-level sentiment measured in the days and weeks following an announcement. In addition, we assume that tariffs and geopolitical rivalry in high-tech sectors affect virtually every firm, albeit to varying degrees. This universal effect follows the exogeneity assumption, allowing us to both isolate our explanatory mechanisms from the macro landscape and ensure theoretical coherence.

Geoeconomic Centrality: Theoretical Framework, and Application to Tariffs

We extend the geoeconomic framework developed by Clayton, Maggiori, and Schreger (2023) by introducing firm-level heterogeneity in compliance capacity into what has been a hegemon-centered account. Clayton et al. characterize hegemonic power along two dimensions. First, *Micro-Power* captures how much a hegemon can extract from a targeted firm or country. It is built by coordinating punishment across economic relationships so that noncompliance in any single one triggers retaliation across many. This coordination therefore relaxes the target's incentive constraints and makes larger demands enforceable. Second, *Macro-Power* captures the social value the hegemon extracts by using its Micro-

Power to reshape the global environment in its favor. Because sectors are linked through input-output networks, demanded actions in one sector propagate through the wider economy, amplifying the hegemon's returns. Within this framwork, the high-tech sector is a good empirical setting for our work, as it is central to the exercise of both Micro and Macro-Power for the United States following the Clayton, Maggiori, and Schreger (2023) framework. Accordingly, American tech giants control infrastructure, software, and inputs on which other countries and firms depend on, so they occupy a unique position in the global economy. If the U.S. government could successfully coordinate joint threat against an enemy utilizing these firms, it could meaningfully translate that coercion into strategic gains.

In this framework, firms voluntarily comply with the hegemon's demands whenever joint threats generate enough continuation value to make compliance individually rational, and firms appear as relatively uniform recipients of coordinated joint threats. We argue that this premise misses an important part of the picture, especially in the high-tech industry: firms differ in their capacity to resist, delay, or coordinate with others for joint threats, and this variation is dictated by their position in strategic supply networks. Geoeconomically central firms possess voice that peripheral firms lack. At the Micro level, geoeconomically central firms weaken the credibility of joint threats because their structural position generates countervailing leverage the hegemon must accommodate; punishing them imposes costs on the hegemon's own network. At the Macro level, geoeconomically central firms shape how demanded actions propagate through the network of production, as they can absorb, redirect, or amplify these effects depending on their own strategic interests. Our extension therefore treats firms as active players in the geoe-

conomic apparatus whose preferences can diverge from those of the state, and identifies the two factors - chokepoint control and mobilizability - that determine when there is a divergence in their response to tariffs.

Building on the framework outlined above, we theorize two mechanisms underlying firms' geoeconomic centrality under great-power rivalry. The first, *chokepoint control*, captures the magnitude of damage a firm can inflict on adversaries' economies when it is excluded from geoeconomic relationships, or the firm's centrality to the hegemon's Macro-Power. In this sense, firms that supply non-substitutable inputs, occupy chokepoints in adversaries' supply chains, or hold technological dependencies that adversaries cannot readily replace impose high potential costs on adversaries. The second, *mobilizability*, captures how easy it is for the state to compel a firm's compliance with geoeconomic strategy and effectively utilize the firm's goods as instruments of coercion, which corresponds to the hegemon's Micro-Power over the firm itself. These two mechanisms are conceptually distinct but empirically interactive, and variation in firm support for tariffs, we argue, tracks these combinations.

On the chokepoint control mechanism, we expect that firms occupying more upstream positions in global value chains (GVCs) are less negative about tariffs than downstream ones. Following Antràs et al. (2012), we define upstreamness as how far a firm's economically important outputs are from the final use so more upstream firms control critical inputs, technologies, or chokepoints that are unsubstitutable (Fishman 2025; Farrell and Newman 2019; Clayton, Maggiori, and Schreger 2024), eventually more essential for the state's Macro-Power. Firms such as advanced semiconductor manufacturers, lithography toolmakers, and EDA software providers as well as cryogenics for quantum computing

occupy central nodes in production networks, where their outputs are essential for a wide range of downstream applications and their substitutes are rare. Because of that, these firms are highly valuable to states during geopolitical rivalry: when the state can successfully coordinate these firms for its geoeconomic policy, doing so is economically destructive for its adversaries due to the ripple effect caused by upstream disruption (Acemoglu et al. 2012; Gabaix 2011; Ojha et al. 2018). In this regard, controlling upstream firms at chokepoints strengthens a state's coercive capacity in international markets through propagated shocks to the whole global production networks (Becko and O'Connor 2024; Clayton, Maggiori, and Schreger 2023), meaning upstream firms are geoeconomically central.

Additionally, upstream firms have a different cost structure than downstream ones. These firms are often less dependent on imported inputs or intermediate goods for their production; when there is a disruption in the supply chain, they are therefore less vulnerable to such a shock (Baqaee and Rubbo 2023; Huneeus 2018), since they bear less direct cost than downstream assemblers (Johnson 2018; Antràs et al. 2012). This distinct feature of upstream firms has twofold implications. First, with their occupation of strategic nodes in the global supply chain, they enjoy a privileged position in the geoeconomic apparatus: the state depends heavily on the cooperation of these firms to effectively project its geoeconomic power abroad. Consequently, governments have strong incentives to offer the most upstream firms extra benefits, further cementing state power over strategic bottlenecks (Becko and O'Connor 2024). Second, because of their lower dependence on low-cost global sourcing, upstream firms are less vulnerable to tariffs or any geoeconomic tools that generate shocks to the supply of inputs and intermediates.

This lower exposure to negative externalities caused by tariffs means upstream companies are less likely to oppose such regimes. Even when they bear the brunt of tariff costs, these costs could potentially be offset by government support, as mentioned previously. Taken together, upstream firms have weaker incentives than midstream or downstream ones to vehemently oppose tariffs.

Nevertheless, going beyond the part of the literature predicting that upstream firms collectively support, or are indifferent toward tariffs, we expect that within high-tech or strategic sectors, the effect of upstreamness varies greatly among upstream firms, compared to non-strategic or traditional ones. This pattern is well-documented in American trade history: American upstream producers in traditional industries during the 19th and early 20th centuries, notably agriculture and steel manufacturers, collectively and staunchly supported protectionist measures (Schattschneider 1935; Gourevitch 1986; Irwin 2017). Their homogeneous behavior is well-explained by positional logic: upstream producers within a given sector shared the same competitive exposure and the same relationship with the state, making upstreamness a reliable predictor of protectionist tolerance. In the high-tech sector, this homogeneity is less clear, as geoeconomic factors also govern firms' actions, in parallel with their supply-chain position: firms controlling genuine chokepoints attract the state's special treatment and become more moderate, while less critical upstream firms do not receive equivalent treatment and remain more opposed. The result is that upstreamness alone is a weaker and noisier predictor of tariff attitudes within the high-tech setting than in traditional industries, because geoeconomic heterogeneity among upstream firms may attenuate the average effect.

Hypothesis 1: More upstream firms are less negative towards tariffs than

lower-stream ones.

From a geoeconomic perspective, a firm's mobilizability should shape its tariff negativity. Importantly, mobilizability is not simply about how much a geoeconomic tool could hurt a firm. Instead, it is about whether such an instrument can effectively engage the firm in the first place and be sustainably enforced afterward (Blackwill and Harris 2016; Luttwak 1990). Geopolitical and geoeconomic instruments, including tariffs, are designed to inflict as much damage on the target as possible, so they must be imposed on goods that are either tangible or strategically valuable, or both (Mulder 2022). Accordingly, these instruments usually target goods that are physically tangible and quantifiable, given their tractability through cross-border monitoring, which strengthens the state's Micro-Power via easier enforcement. This is the case with high-tech companies: those selling hardware products are easier for the state to mobilize for economic coercion and enforce punishment for violation, and have more exposure to tariffs (Osgood 2018), given the nature of their goods and their position in the global high-tech industry.[3] Granted, these firms could receive favorable treatment from the state, but the costs of sweeping tariffs could exceed the expected benefits of favorable treatment when firms are mobilized.

The logic is as follows. Hardware firms, particularly downstream and midstream ones, import substantial foreign inputs and intermediate goods, so under growing protectionism, these companies could suffer substantial losses. Digital or software-driven firms, on the other hand, sell intangible products that face fewer constraints under the same tariff regimes, as they do not have to cross customs borders. Consequently, we believe that hardware-intensive firms would be more negative toward tariffs than software ones.

3. As of 2026, most high-tech products subject to export controls or sanctions by the United States are physical or hardware-linked. See the Bureau of Industry and Security's Commerce Control List: https://www.bis.gov/regulations/ear/interactive-commerce-control-list?isExpanded=&category=&keyword=

Moreover, because of physical and capital constraints, as well as the extreme degree of specialization in the high-tech value chain, tangible-goods providers usually offer more product differentiation, or are less substitutable, than digitally-driven ones. They are thus be more likely to support for open trade, as shown by recent work that firms that are productive exporters are more likely to lobby for tariff reductions, especially when their products are sufficiently low-substitutable, or highly differentiated (Kim 2017; Osgood 2017, 2016; Baccini and Dür 2018).

Within hardware companies, upstream firms differ from downstream or midstream ones in their tolerance for tariffs. These companies are at the intersection of the state's Micro and Macro-Power: they are both easy to mobilize for joint threats and critical to global supply chains. As argued previously, the successful mobilization of upstream firms is usually more destructive to adversaries, and this is particularly true of American hardware providers. Given their dominance in critical goods, they occupy chokepoints in global technology and defense, so forming a joint threat from these firms against an adversary would be highly damaging to its economy. Combined with the tangibility of their hardware products, these firms are indispensable for the state to accumulate its Micro-Power that is necessary to project Macro-Power and reap its benefits through supply shocks cascading from upstream to the whole global economy (Farrell and Newman 2019; Clayton, Maggiori, and Schreger 2023). These firms are also extremely capital-intensive, so they cannot easily relocate from targeted markets following tariffs, meaning tariff costs are real, but so is the collateral damage to the imposing state due to their criticality to both the home and enemy economies. Thanks to their occupation of economic chokepoints, upstream hardware firms are therefore likely to receive some form of

special treatment from the government, which could offset the losses they incur from tariffs.[4] The implication is that upstream hardware firms have greater tolerance for tariffs, making them less opposed than their downstream counterparts.

Hypothesis 2: Hardware firms are more negative towards tariffs than software ones.

Hypothesis 2a: Upstream hardware firms are less negative towards tariffs than downstream ones.

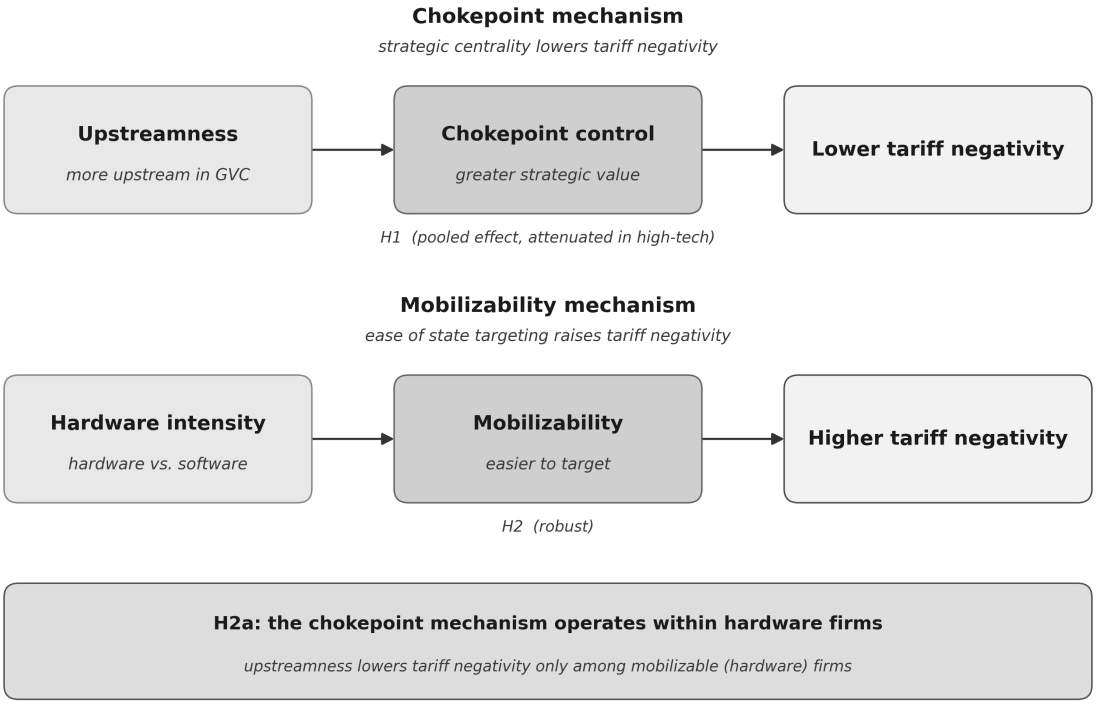


Figure 1: Theoretical Mechanisms

4. Indeed, some of the largest American chipmakers have received significant packages from the CHIPS & Science Act. For example, Intel was awarded $8.5 billion in grants and $11 billion in loans for projects in Arizona, New Mexico, Ohio, and Oregon; GlobalFoundries was awarded $1.5 billion to expand manufacturing in New York and Vermont, specifically for automotive and defense technologies.

Research Design

Textual Data

In order to measure tariff negativity at the firm level, we need sources of firm-level text. Earnings calls are a valuable source of textual information because they allow market participants to pose questions of their own choosing to firm leadership. Although the calls open with prepared remarks from management, the subsequent Q&A session allows participants to raise the issues they find most pressing, which is well-suited to our study of firm-level opinion about tariffs. With the emergence of text analysis and large language models (LLMs), there is a proliferation of studies using earnings calls and other sources of corporate text to examine firm-specific dynamics (Hassan et al. 2024; Clayton et al. 2025; Heinrichs, Park, and Soltes 2019; Bochkay, Hales, and Chava 2020). We use the Capital IQ Transcript data to acquire all earnings-call transcripts of the 65 largest American high-tech firms by market capitalization (ranked by Fortune 500) from the first quarter of 2024 to the first quarter of 2026. This is a highly reliable source: the transcripts are well-structured and cover the full set of sample firms with no missing documents. The dataset is widely used in academic and industry research.

We focus only on American high-tech firms for several reasons. First, high-tech sectors are at the forefront of great-power competition between the United States and China, meaning they are heavily exposed to coercive measures from both sides, including tariffs. American tech companies are heavily dependent on manufacturing and supply chains running through the European Union, China, and Taiwan, so they would be hit hard by tariffs. The head of the Consumer Technology Association warned that tariffs would

"drive inflation" and "kill jobs," risking a recession in the United States (Shapero 2025). Second, the high-tech sector is a hard test for our theory. Because of their dominant role in the global high-tech supply chain, American tech firms are likely to be mobilized by the state, following the geoeconomic logic, and play a critical role in the implementation of American tariff policy. Because of that, almost all high-tech firms enjoy some degree of preferential state treatment. If our results are significant even within this already-privileged sector, the effects of our proposed mechanisms must be substantively strong. Third, because American firms are globally dominant, studying how they respond to tariffs is foundational to understanding the political-economic foundations of modern American geoeconomics, given the intensification of US-China high-tech rivalry and the US weaponization of technological interdependence.

Methodology

As LLMs have become more advanced, a number of studies have shown that these models can perform text annotation, detection, classification, and coding reliably (Gilardi, Alizadeh, and Kubli 2023; Le Mens and Gallego 2025; Umansky et al. 2026). We use LLMs to extract our key variables. We perform large-scale inference on a textual corpus of 486 earnings calls using frontier proprietary LLMs. For our baseline results, we code the earnings-call transcripts with OpenAI's GPT-4o models in a two-step pipeline. In the first step, GPT-4o-mini reads each transcript (approximately 10,000 words) and extracts the relevant passages verbatim: statements on trade policy, trade-adjacent language, and descriptions of the firm's business model. This compresses each transcript to roughly 1,000 words of informative excerpts. In the second step, GPT-4o reads the

extracted passages and scores each call on tariff negativity, upstreamness, and the other latent trade-environment measures. We assign the two steps to different models for cost efficiency. Extraction is a simple task: the instructions are clear, and the output consists of verbatim quotes, which GPT-4o-mini handles reliably.[5] Scoring is more demanding, as it requires nuanced interpretation of tone and context, followed by calibration against a coding rubric so we use 4o for this more complicated task.

We adopt a two-step coding architecture rather than a single-pass approach for both computational and methodological reasons. On the computational side, earnings-call transcripts average approximately 10,000 words, and submitting the full text of each transcript directly to the scoring model would impose substantial token costs across 486 observations while risking truncation at context-window limits. More importantly, the two-step design offers a methodological advantage grounded in the structure of the underlying text. Earnings-call transcripts are overwhelmingly composed of content unrelated to trade policy: quarterly revenue figures, product roadmaps, segment-level guidance, and analyst questions about margins, backlog, and capital allocation. Trade-relevant language, when present, typically constitutes fewer than 1,000 words embedded within this larger corpus. Directing a scoring model to evaluate the full transcript risks diluting signal with noise, as the model must simultaneously identify relevant content and assess its valence and intensity. Separating these tasks allows each model to specialize: the extraction model identifies and isolates trade-relevant passages without evaluating their content, while the scoring model evaluates those passages without the distraction of irrelevant material. This procedure mirrors standard practice in manual content analysis,

5. See the Appendix for the extracted snippets.

in which a research assistant first identifies and excerpts relevant text segments before a trained coder applies the coding scheme to those segments (Krippendorff 2018). The decomposition reduces coder burden, minimizes context-dependent errors, and ensures that the scoring model's attention is concentrated on the substantively relevant portions of each document. We provide the full prompt in the Appendix.

Empirical Strategy

There are three distinct identification threats to our causal inference. First and most importantly, firm-level GVC characteristics are endogenous and may correlate with unobservable determinants of expressed tariff attitudes. Firms do not randomly select into physical or digital value chains, and the characteristics that determine GVC position may independently shape how firms communicate about trade policy. A second threat is anticipation: Trump won the election in November 2024, and tariff threats were among his central campaign promises. Firms may have begun adjusting their rhetoric before the tariffs took effect. If firms started expressing more negativity in anticipation, our post-treatment effect might capture strategic repositioning rather than a response to the actual policy. A third issue is that earnings-call rhetoric may reflect strategic communication rather than sincere attitudes. Firms communicate with multiple audiences simultaneously, from investors, regulators, policymakers, to competitors, and may calibrate their expressed tariff attitudes accordingly.

We employ a difference-in-differences (DiD) design, leveraging the Liberation Day tariffs (April 2, 2025) as a time cut-off between pre and post-treatment period. DiD is among the most widely used quasi-experimental designs in causal inference, with early

applications of its logic dating at least to Snow's (1855) analysis of the 1854 London cholera outbreak; it compares the change in outcomes, before versus after a shock, across groups that differ in their exposure to it. Because the tariff shock does not discriminate among firms but affect them differently, we identify its effects from differential exposure—hardware versus digital firms, and upstream versus downstream firms—rather than from a stark treated-versus-untreated comparison. This differential-exposure approach is standard for studying the effect of a policy or other exogenous shock on units that vary in their exposure to it (Autor, Dorn, and Hanson 2016; Pierce and Schott 2016). Our specification is the following:

$$Y_{it} = \alpha_i + \gamma_t + \beta_1(H_i * P_t) + \beta_2(U_i * H_i * P_t) + \varepsilon_{it} \tag{1}$$

where Y_{it} is tariff negativity for firm i in quarter t, scaled from 0 (no mention or neutral framing) to 1 (severe negativity). α_i denotes firm fixed effects, which absorb all time-invariant firm characteristics including industry membership, business model, organizational culture, and baseline political orientation. γ_t denotes quarterly (year $\times$ quarter) fixed effects, which absorb common macroeconomic conditions and policy shocks affecting all firms in a given quarter. H_i is a binary indicator equal to 1 for firms whose products take physical form and cross customs borders (semiconductors, hardware systems, defense equipment, and networking hardware) and 0 for firms whose value creation is primarily digital (software, cloud services, internet platforms, and IT services). P_t is a binary indicator equal to 1 for quarters beginning in Q2 2025, immediately following the Liberation Day tariff announcement of April 2, 2025. U_i is the standardized pre-treatment mean of the LLM-coded upstreamness score, ranging from -1 (pure downstream, consumer-

facing) to +1 (pure upstream, inputs supplier), averaged across pre-treatment quarters to construct a time-invariant firm characteristic.[6] We adopt a parsimonious approach, do not include additional time-varying firm-level controls, as these are plausibly affected by the tariff shock and would constitute post-treatment bias on the causal path between tariff imposition and expressed attitudes (Angrist and Pischke 2009; Montgomery, Nyhan, and Torres 2018). This follows a practice for differential-exposure designs in the literature of economics, which do not include any additional control covariates except unit and time fixed-effects (Acemoglu and Finkelstein 2008; Nakamura and Steinsson 2014). Standard errors are clustered at the firm level to account for serial correlation within firms across quarters (Bertrand, Duflo, and Mullainathan 2004). Turning to the concern about strategic communication, we believe this threat is attenuated in our context. Corporate signaling operates through private information: it generates value when firms know more about their exposure than investors do. Under the Liberation Day regime, tariff rates and sector-level exposure were publicly observable, eliminating the informational asymmetry that makes investor signaling valuable. What firms communicated under these conditions is therefore more likely to reflect genuine political preferences than investor-relations strategy, particularly for upstream hardware firms facing the strongest incentives to seek more favorable policy adjustment.

Several lower-order interaction terms are implied by the triple interaction in our specification: $Hardware_i$, $Upstream_i$, $Upstream_i \times Hardware_i$, and $Post_t$ (Brambor, Clark, and Golder 2006). The first three are time-invariant and therefore absorbed by firm fixed effects; the fourth is absorbed by calendar-quarter fixed effects. The remaining

6. We include our verbatim LLM prompt in the Appendix.

lower-order term, $Upstream_i \times Post_t$, is omitted from our preferred specification on theoretical grounds: our framework predicts that upstreamness affects tariff attitudes primarily through customs exposure, which operates only on firms whose products take physical form and cross customs borders. A general upstreamness gradient extending to digital firms is not implied by our argument, and the within-digital point estimate is correspondingly small and not statistically distinguishable from zero (Table 2). Because $Upstream_i$ and $Hardware_i$ are correlated at $\rho = 0.53$ in our sample, including $Upstream_i \times Post_t$ splits the same within-hardware gradient into two weakly identified components ($\hat{\beta}_{U\times P} = -0.036$, $p = 0.34$; $\hat{\beta}_{U\times H\times P} = -0.046$, $p = 0.31$); their sum, the within-hardware total gradient, equals -0.082 (SE $= 0.025$, $p = 0.001$), identical to our preferred specification.

Results

Before presenting the main findings, we first verify that the parallel trends assumption holds in our design. This is key to causal identification under the DiD, because the assumption justifies using the control group's pre-to-post change as the counterfactual trend for the treated group. If parallel trends hold, any additional post-treatment divergence between treated and control groups can be interpreted as the causal effect of the treatment. As a first cut, we conduct a Wald test of the joint null hypothesis that all pre-treatment event-study coefficients equal zero; we fail to reject the null, meaning there is no evidence contradicting the parallel trends assumption. We find further corroborating evidence using an event study. As shown in Figures 2 and 3, the pre-treatment coefficients are individually close to zero and statistically insignificant, as well as jointly

indistinguishable from zero.

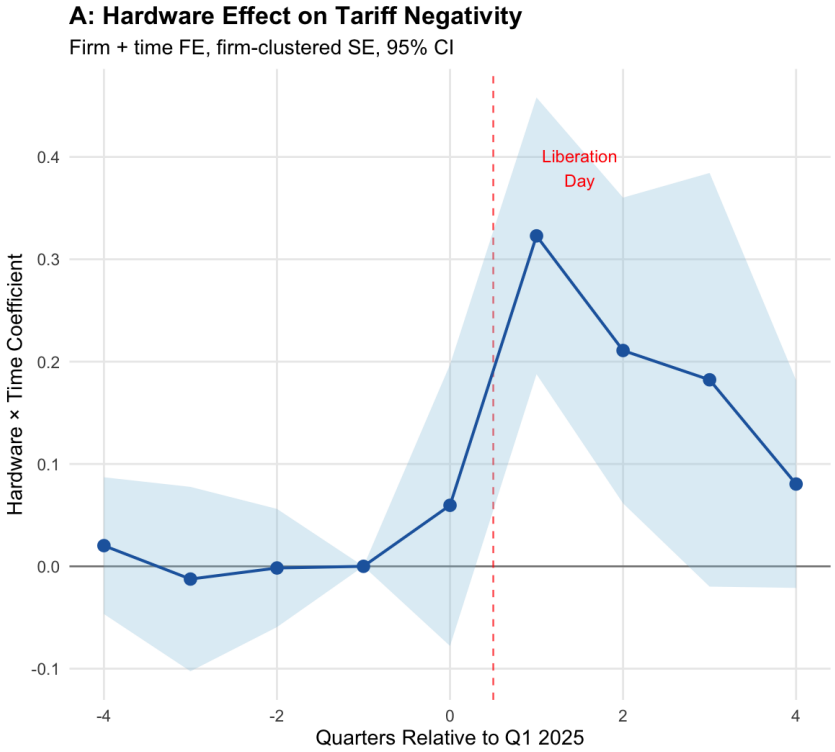


Figure 2: Tariff Negativity of Hardware Firms

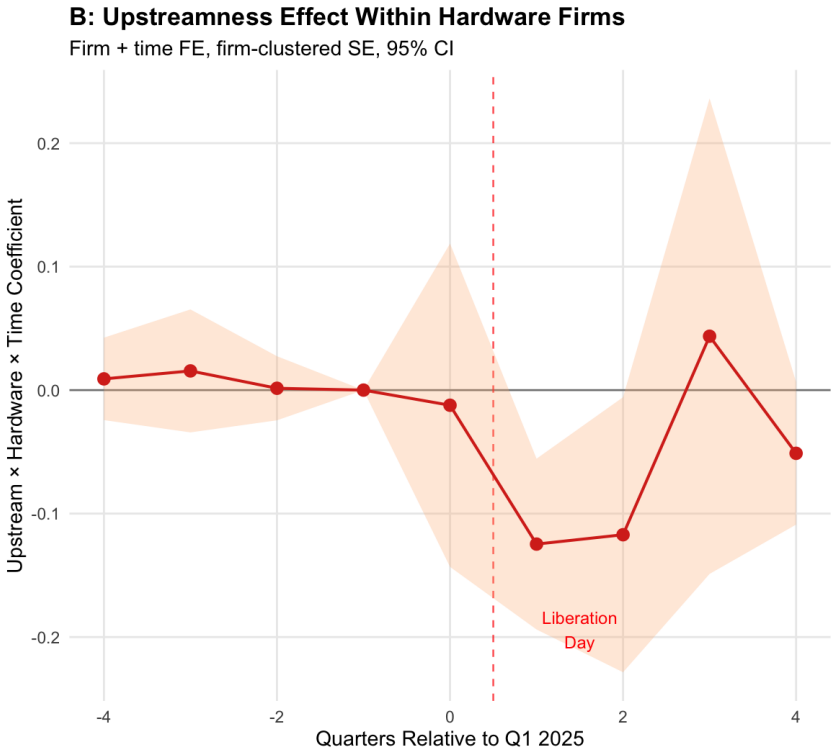


Figure 3: Effects of Upstreamness on Hardware Firms' Negativity

In addition, we have strong descriptive evidence that the Liberation Day tariffs dramatically altered the salience of trade policy in firm communications. Figure 4 plots the share of firm-quarter observations in which the LLM coding procedure detects any tariff-related discussion (i.e., observations with a tariff negativity score above zero) by calendar quarter. Throughout 2024, fewer than 5% of earnings-call transcripts in our sample contain any tariff-related discussion. This share rises modestly to 22% in Q1 2025 as Trump's tariff agenda became increasingly central to the political conversation following his return to office. Following Liberation Day on April 2, 2025, tariff discussion appears in 75% of Q2 2025 transcripts, which is a 15-fold increase from the 2024 baseline. The share declines through Q3 and Q4 2025 as firms adapt to the new policy environment, then drops sharply to 25% in Q1 2026 following the Supreme Court's in-

validation of tariffs imposed under the International Emergency Economic Powers Act (IEEPA) of 1977.[7] The trajectory closely tracks specific policy events, from pre-election baseline, anticipation, treatment onset, to treatment removal following the SCOTUS' decision, providing strong descriptive evidence that the Liberation Day shock changed how firms communicate about trade policy. The close correspondence between our LLM-coded salience measure and the timeline of independently verifiable policy events further validates the construct validity of the coding procedure: the measure tracks the actual policy environment rather than capturing unrelated linguistic variation.

The convergence following the Court's invalidation of tariffs is particularly telling. Convergence after treatment removal is a more demanding test than divergence after imposition: a confound coinciding with Liberation Day could generate the initial divergence, but would have to independently reverse at the Supreme Court's decision to generate the subsequent convergence. The convergence toward the negativity baseline right when the tariffs are struck down is difficult to reconcile with any such confound, and points to the tariffs themselves as the driver. We interpret this cautiously, nevertheless, as it rests on a single post-invalidation quarter, and the removal was contested by continued efforts from the Trump administration to keep its tariff regimes.

7. On February 20, 2026, the U.S. Supreme Court overturned those tariffs in *Learning Resources, Inc. v. Trump* and *Trump v. V.O.S. Selections, Inc.*, ruling that IEEPA does not grant the power to set tariffs.

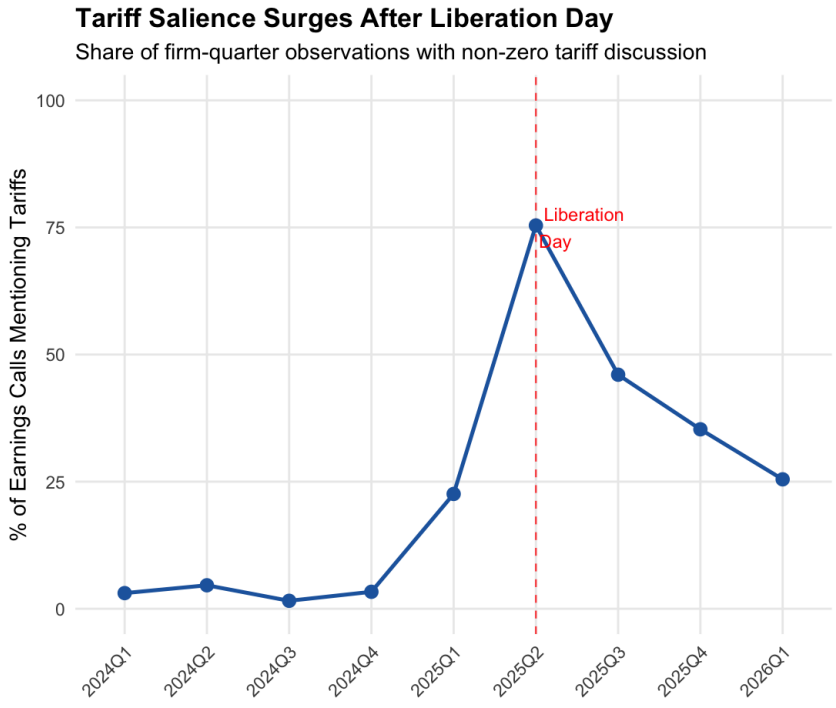


Figure 4: Tariff Salience Across Quarters

Our main results are in Table 1. Overall, they provide strong support for H2 and H2a. Hardware firms became more negative toward tariffs than software firms following Liberation Day by 0.139 points (Model 2), and within hardware firms, a one-standard-deviation increase in upstreamness reduces this effect by 0.082 points (Model 3). The level effect is robust under cluster-robust inference at the 1% level, and the within-hardware gradient remains significant at conventional levels under both CR2 cluster-robust standard errors with Satterthwaite degrees of freedom ($p = 0.034$) (Bell and McCaffrey 2002; Pustejovsky and Tipton 2018) and the restricted wild cluster bootstrap with Webb 6-point weights ($B = 9{,}999$, $p = 0.031$) (Cameron, Gelbach, and Miller 2008; MacKinnon and Webb 2018). Our result is also robust to leave-one-cluster-out validation: across all 32 specifications that drop each hardware firm in turn, the coefficient remains significant at the 1% level, with point estimates in $[-0.097, -0.064]$. In contrast, the

coefficient on Upstream × Post in Model 1 is small and not statistically significant, consistent with our framework: as anticipated (see Empirical Strategy), this null result reflects the customs-exposure channel, which does not apply to digital firms. The gradient should therefore be concentrated among customs-exposed firms, which is what Models 2 and 3 demonstrate directly.

Table 1: Difference-in-Differences Estimates of Tariff Negativity

	(1) Baseline	(2) H2	(3) H2 + H2a
Upstream × Post	−0.002		
	(0.026)		
Hardware × Post		0.139***	0.184***
		(0.044)	(0.042)
Upstream × Hardware × Post			−0.082***
			(0.025)
Firm FEs	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes
Observations	486	486	486

Notes: Standard errors clustered at the firm level in parentheses. The within-hardware gradient (H2a) remains significant at conventional levels under both CR2 cluster-robust standard errors with Satterthwaite degrees of freedom ($p = 0.034$) and the restricted wild cluster bootstrap with Webb weights ($B = 9{,}999$, $p = 0.031$).
$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

To test this interpretation, we estimate the upstream effect separately within digital and hardware firms (Table 2). Among digital firms, upstreamness has no effect on tariff negativity, consistent with the expectation that GVC position is largely irrelevant when products do not cross customs borders. Among hardware firms, the upstream effect is negative and significant. The split-sample analysis is inherently more sensitive than the full-sample interaction because it re-standardizes upstreamness within each subsample and operates on a smaller N. The role of the split-sample analysis here is supplementary: it directly demonstrates that upstreamness has no effect among digital firms, and that the

effect persists among hardware firms, validating our exclusion of the $Upstream_i \times Post_t$ interaction term.

Table 2: Split-Sample Analysis: Upstream Effect by Hardware Status

	(1) Digital Firms	(2) Hardware Firms
Upstream × Post	−0.029	−0.070***
	(0.032)	(0.022)
Firm FEs	Yes	Yes
Quarter FEs	Yes	Yes
Observations	243	243

Notes: Standard errors are clustered at the firm level in parentheses. Upstreamness is standardized within each subsample, so coefficients are on within-group scales and are not directly comparable in magnitude to the triple interaction in Table 1.
$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

To further probe our parallel trends assumption, we calculate the raw mean of tariff negativity across the full sample period. The absence of pre-treatment differentiation between hardware and software firms would be consistent with the parallel trends assumption. Figure 5 illustrates this clearly: before Q1 2025, hardware and digital firms display nearly identical, and statistically insignificant, attitudes toward tariffs. Immediately following Q1 2025, both types of firms become more pessimistic about tariffs, but hardware firms become noticeably more negative afterward, which can be explained by the anticipation effect: firms across both categories began discussing tariffs more frequently as Trump's hardline trade rhetoric intensified, and expressed growing uncertainty about the effects of tariffs on their businesses. The differential response emerges only in Q2 2025, immediately following Liberation Day, when hardware firms diverge sharply upward while digital firms increase only modestly. Interestingly, we also see a steep decrease in hardware firms' negativity in Q1 2026, right after the Supreme Court struck down Trump's tariffs. This reconvergence of the two groups after the Supreme Court's

ruling strengthens our causal finding: the cleavage in negativity disappears immediately after the treatment is removed.

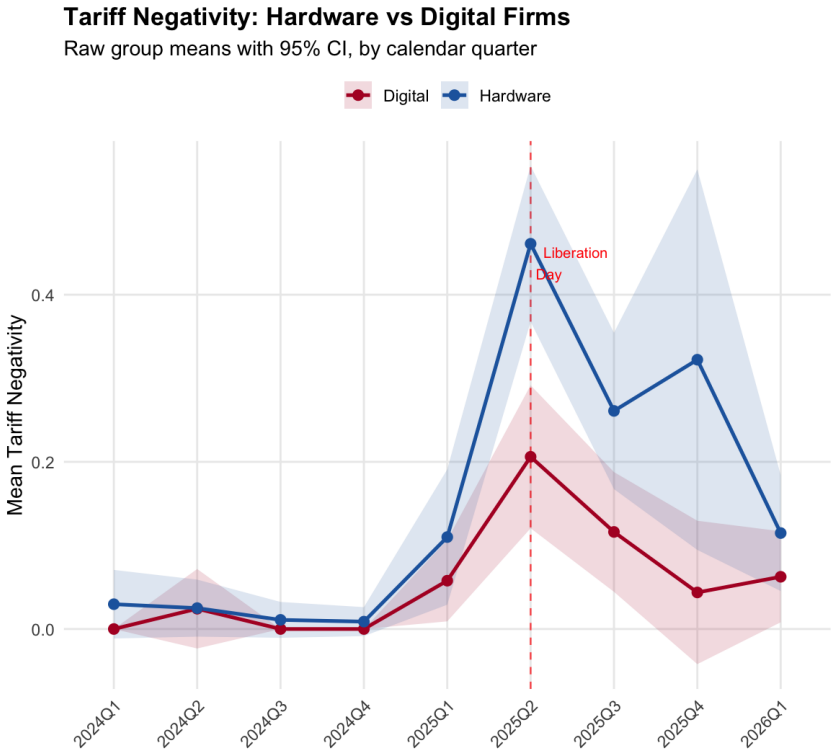


Figure 5: Mean Tariff Negativity

Preference Validation: Tariff-Related Lobbying Patterns

To examine whether the attitude toward tariffs extracted from earnings-call transcripts is consistent with business preferences, and not merely an artifact of strategic communication, we look at the pattern of how firms lobby on trade- and tariff-related issues, and which government bodies they target. Lobbying is one of the most important political channels through which firms advocate for their preferences (Grossman and Helpman 1994; Woll 2008; Kim 2017; Baumgartner et al. 2009; Schnakenberg 2017). Therefore, if lobbying patterns are consistent with tariff salience, it is plausible to say that firms

reveal their true preferences in their calls.

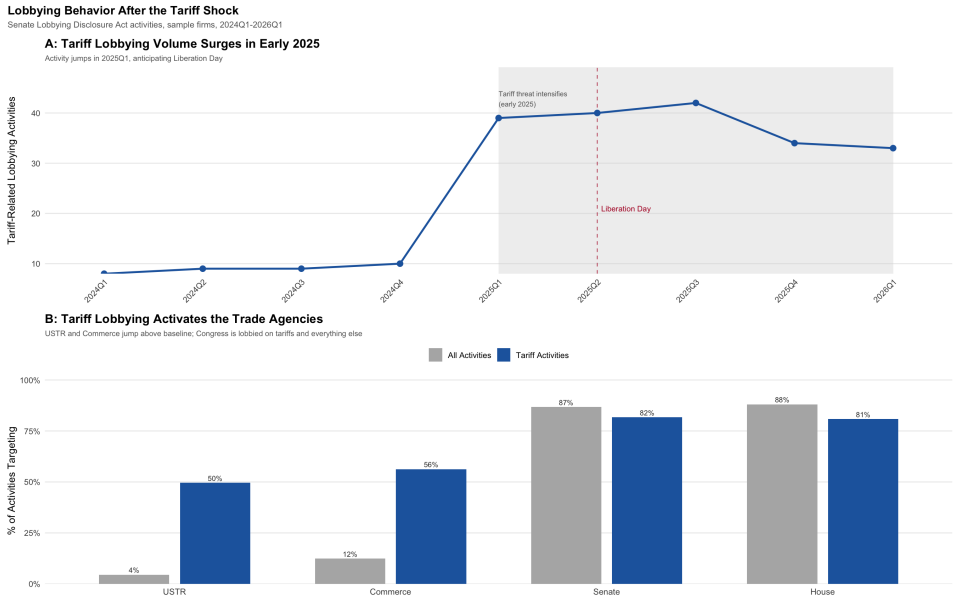


Figure 6: Tariff-Related Lobbying Behavior

Figure 6, Panel A, shows a consistent rise in tariff-related lobbying that tracks the growing concern about tariffs in firms' earnings calls. There is a surge in lobbying activity in early 2025, shortly after Trump took office, and the trend has been stable since then. Although firms could in principle lobby for *more* tariffs, which would run counter to our argument, we contend that in the context of the tariffs proposed by the Trump administration, this is unlikely. This tariff regime is considered the most sweeping one in modern U.S. history, and it was expected to have a devastating effect on the American economy and business activity, albeit to varying degrees (York and Durante 2026). It is thus unlikely that firms would lobby *for* such an economically destructive regime. In other words, we can confidently say that the surging lobbying activity since Q1 2025 is *against* tariffs. Taken together, the intensification of tariff-related lobbying activity and the parallel shift in tariff salience and negativity tell a coherent story: firms are not only

showing an unfavorable attitude toward tariffs but making an effort to mitigate the costs to their business. Panel B reinforces this pattern. Relative to other lobbying activity, tariff-related lobbying is more heavily directed toward the United States Trade Representative and the Department of Commerce. This pattern is institutionally consistent, as the 2025 tariffs were imposed and administered by the executive branch, not the legislature. As expected, when firms advocate for their interests, they target the venues that are directly involved in the process. Taken together, our interpretation—that earnings calls reflect genuine, costly political engagement and preferences rather than cheap talk or strategic investor-relations communication—is strongly supported.

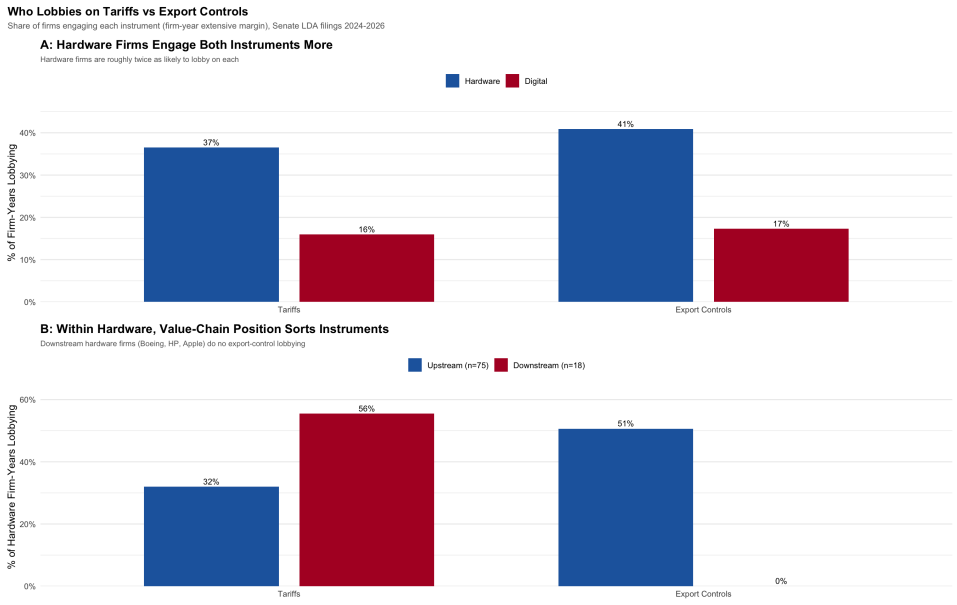


Figure 7: Tariff and Export-Control Lobbying by Firm Type

The lobbying evidence also reproduces the heterogeneity that organizes our main results. Figure 7 disaggregates lobbying by firm type, using the share of firms in each group that engage a given instrument. Hardware firms are roughly twice as likely as digital firms to lobby on both tariffs (37% versus 16%) and export controls (41% versus

17%), illustrating the hardware-digital divide in rhetorical tone toward tariffs and related instruments. This behavioral pattern confirms that the divide is a feature of how firms act, not merely how they talk. Within hardware, value-chain position sorts firms across instruments: downstream producers such as Boeing, HP, and Apple concentrate on tariffs (56% of firm-years) and do not engage with export-control lobbying, while roughly half of upstream semiconductor firms lobby on export controls (51%). This reflects the structure of exposure: downstream firms bear tariff costs on imported goods, whereas upstream firms' principal vulnerability is restrictions on their sales to China. This divergence also provides a behavioral counterpart to the forbearance of upstream firms in our main analysis. That said, the within-hardware comparison rests on a modest number of clearly downstream hardware firms, so we read it as descriptive corroboration rather than an estimated effect.

Mechanism Check: Special Treatment

Under our second hypothesis, we argue that upstream hardware firms' muted stance partly reflects special treatment from the government, owing to their position in the geoeconomic apparatus. An alternative explanation is that they are simply less dependent on foreign inputs, such that cost structure, rather than geoeconomics, drives their tariff concern (Antràs et al. 2012; Alfaro and Chor 2023). We regard this as valid but incomplete. It cannot explain the divergence between Apple and Qualcomm, both heavily dependent on imported inputs (Reinsch and Whitney 2025; Alfaro and Chor 2023): Apple on imported components and assembly, Qualcomm on Taiwan-fabricated chips.[8]

8. Qualcomm's fiscal Q3 10-Q states: "We depend on a limited number of third-party suppliers for the procurement, manufacture, assembly, and testing of our products manufactured in a fabless production model." See: https://s204.q4cdn.com/645488518/files/doc_financials/2026/q3/QCOM-Q3FY26-10-

Tariffs should burden both, yet their attitudes diverge—suggesting that special treatment also plays a role, one whose importance grows as import dependence rises.

To test this, we examine whether upstream products are more likely to receive Section 301 tariff exemptions, using exemption data compiled by Brunk, McDaniel, and Parks (2019) for 2017–2020. Although this precedes our main period, the exemption process - administered by the USTR through a standardized procedure applied uniformly across firms - is the standard channel for trade-remedy relief, so its logic plausibly extends to 2024–2026. We treat these exemptions as evidence of whether the mechanism operates, not as a measure of our treatment period.

The unit of analysis is the HTS-10 product rather than the firm, since exemptions are granted at the product level and aggregating to firms would dilute a product's outcome across the firms selling it. The dependent variable equals one if a product was ever granted an exemption. The key regressor indicates whether the product is upstream (an intermediate input or capital good) versus a consumer good. To capture the cost-structure channel, we control for import exposure: the natural log of average U.S. imports of the product, measured at the eight-digit HTS level, which we treat as a close proxy for each ten-digit product's exposure. We expect the two channels to differ in intensity across import exposure. At low exposure, little is at stake; as exposure rises, nearly all firms have an incentive to seek exemptions, so exposure alone ceases to distinguish products, and upstreamness becomes the decisive differentiator. The effect of upstreamness should thus *grow* with exposure—an interaction that need not be linear. We therefore estimate the conditional marginal effect using the binning estimator of Hainmueller, Mummolo, and

$Q_Final.pdf$.

Xu (2019), which partitions the moderator and estimates the effect within bins without imposing linearity, and we restrict the sample to products with observed import values.

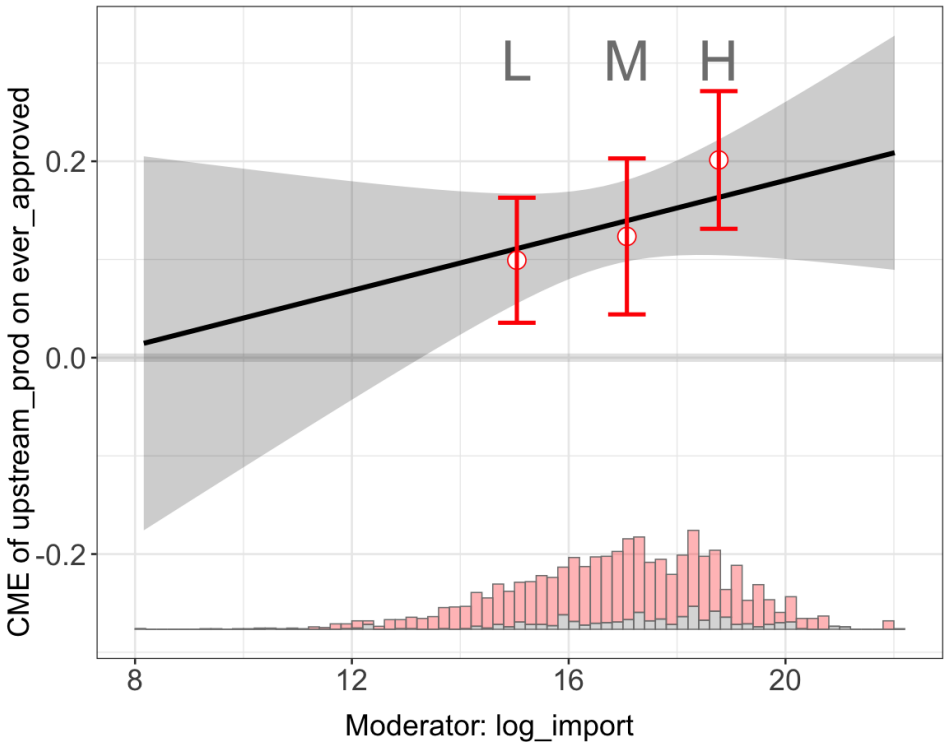


Figure 8: The Conditional Marginal Effects of Being Upstream to The Outcome

Figure 8 supports our argument. The marginal effect of upstreamness on the probability of exemption is positive throughout and rises with import exposure, from roughly 10 to 20 percentage points, with all binned estimates significant across the region of common support. Upstreamness increasingly determines which products obtain relief as exposure grows, and this effect is consistent with the mechanism underlying H2a. Because upstream products more readily secure exemptions, the upstream hardware firms supplying them are more insulated from tariff costs, which accounts for their lower tariff negativity relative to downstream firms. Nevertheless, this evidence is suggestive rather than causal: it establishes that upstream products obtain exemptions at higher rates, not

that these exemptions causally drive firms' expressed attitudes. Importantly, combined with the evidence from the lobbying pattern, we could rule out the explanation that upstream hardware firms are less negative about tariffs because they are less dependent on imports so they are not attentive to tariffs.

Discussion

Our results point to a coherent story: under rising tension between the United States and China, American high-tech firms diverge in their perception of tariffs, and this divergence depends on their principal product category and their position in the supply chain. Geoeconomically, the former determines how easy it is to mobilize a firm; the latter is about whether a firm is strategically substitutable, and the two interact in shaping firms' attitudes and behavior. Our theory applies not only to high-tech firms but also to those in other critical sectors. This is clearly demonstrated in the financial sector. The United States has for decades leveraged its influence over the Society for Worldwide Interbank Financial Telecommunication (SWIFT) to sanction Iran and Russia.[9] Given its irreplaceability in the financial system and the centrality of its product, it comes as no surprise that the organization and its members are integral to the American geoeconomic apparatus (Fishman 2025; Cipriani, Goldberg, and La Spada 2023; Bapat et al. 2024).

The cleavage among American high-tech firms has profound implications for policymakers. First and most importantly, although tariffs are designed as an instrument to achieve foreign economic objectives, their effectiveness is questionable, given the potential for resistance from firms that are expected to follow the flag. Accordingly, variation in

9. Some of SWIFT's key members, including Goldman Sachs and JPMorgan, are ambiguous at best about tariffs. See: https://www.cnbc.com/2025/04/11/jpmorgan-chase-jpm-earnings-q1-2025.html; https://www.cnbc.com/2025/04/14/goldman-sachs-gs-earnings-q1-2025.html

negativity among American hardware firms may undermine the full implementation of tariffs, as the firms most affected attempt to delay or block them, individually or collectively, much as firms have throughout American trade history (Irwin 2017; Milner 1988; Woll 2008). This pushback could eventually undermine American geoeconomic power, given the global dominance of American tech firms, particularly in hardware. Additionally, tariffs might cause collateral damage to the American economy by hurting its tech companies. American high-tech firms have strong integration into the Chinese market and industrial base, and they are indispensable to both the American economy and the global economy (Hogg, Cimerman, and Johnson 2025; Wolf and Terrell 2016). If China continues to escalate retaliation in response to American tariffs and broader geoeconomic pressure, these companies would lose revenue that is central to their research and development activities (Leahy, Hu, and Sevastopulo 2026). To the extent that these measures impose sustained revenue costs on American hardware firms, they create a risk of cumulative erosion of competitive position and innovation capacity over the medium term. Paradoxically, the instrument intended to strengthen American economic security may, through retaliation and integration effects, weaken the firms that constitute the backbone of American economic power.

Conclusion

The divergence of American tech companies in their response to tariffs is both theoretically puzzling and empirically under-tested, as existing frameworks have not fully explained this phenomenon. Through the geoeconomic lens, we offer a novel explanation for firm behavior under tariffs, and more broadly under economic tension. Although we

do not claim that this is the only factor accounting for the cleavage, we believe that our geoeconomic argument is valuable to the IPE literature, and, to some extent, to international security. Geoeconomics has emerged as a line of research in IPE over the past decade, but very few studies examine tariffs as a geoeconomic instrument or ask how firms matter geoeconomically from this perspective. Our findings therefore shed light on tariffs and the active role of firms as geoeconomic players, which challenges the strand of literature assuming that firms passively follow the flag and lack geoeconomic preferences of their own. Our findings are also relevant to the international security literature. Economics is now a critical dimension of national security, and tariffs themselves have increasingly been justified on national-security grounds. As a result, large firms, particularly high-tech ones, are now playing a more significant role in the economic security apparatus, thanks to their resources, expertise, and technological dominance. Understanding how large technology firms differ in their perception of tariffs is thus necessary to examine their influence on the making and implementation of security policy.

Notwithstanding its contributions, this paper has some limitations. Due to data availability, we focus only on American companies, which makes our results less generalizable relative to a cross-national research design. Though leading high-tech companies are relatively homogeneous in their incentive structure and supply-chain characteristics, future studies can expand the analysis to non-American tech companies to stress-test the geoeconomic framework. Next, while the LLM-coded upstreamness measure works well for our design, we acknowledge that using existing firm-level supply-chain data would be more standardized and would allow us to capture additional features, such as buyer-supplier relationships and input-output tables. Nevertheless, our research paves the way

for other studies using the same LLM-based approach to extract more fine-grained firm-level supply-chain information and triangulate it with other established data sources to verify how well LLMs perform at extracting firm-level features. Relatedly, our empirical setting is, to some extent, idiosyncratic to American trade policy: the tariffs applied unusually broad rates across many trading partners simultaneously, and reflected the distinctive trade ideology of a particular administration. Granted, this setting offers methodological advantages: its unusually broad scope, its disruptiveness, and its later judicial invalidation together provide a cleaner identification strategy than more gradual tariff implementations would afford. Future research could extend this approach to other geoeconomic instruments, including but not limited to CHIPS Act allocations, export controls, investment screening, and reciprocal foreign measures, as fruitful directions for the geoeconomic framework. In conclusion, tariffs are no longer purely economic instruments, and firms are no longer purely economic actors. Understanding the geoeconomic logic that ties them together is a crucial step toward making sense of the political economy of great-power rivalry.

Declaration of Statement

The authors declare no conflict of interests.

Declaration of the Use of AI

The authors used AI for editorial assistance, generating graph and data analysis assistance. In particular, the author asked Claude Sonnet 5.0, September 2026 to proofread

the text, reformat and debug the LaTex code for the paper, and instructed ChatGPT 5.6 Sol, September 2026, to construct Figure 1. The authors also used Claude Opus 4.7, May 2026, to assist with revising the R code for data analysis and provide a review report for the paper. After using these tools, the authors reviewed and revised all outputs as needed to ensure the final version fully reflects our expertise, voice, and originality. The authors take full responsibility for the content of the published article.

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Online Appendix: Tariffs, Geoeconomics and Political Cleavages

Contents

A. Descriptive Statistics & Validation **2**

A1. Summary Statistics . . . 2
A2. Sample Construction and Firm List . . . 3
A3. Illustrative Tariff Discussion Excerpts . . . 5
A4. Illustrative Business Model Excerpts: Upstream vs. Downstream . . . 6
A5. Cross-Model Reliability: GPT vs. Claude . . . 8
A6. LLM Prompts . . . 11
A7. Human Validation . . . 19
A8. Upstreamness Measurement Validation . . . 21

B. Robustness Tests **22**

B1. Binary Dependent Variable . . . 22
B2. Exclusion of Defense Firms . . . 23
B3. Placebo Test . . . 25
B4. Two-way Clustered Standard Errors: Firm and Calendar-Quarter . . . 27
B5. Alternative DV: Tariff-Related Lobbying . . . 28
B6. Wild-Cluster Bootstrap, CR2 Standard Errors and Leave-One-Out Estimate . . . 32
B7. Specification with 10-K Upstreamness . . . 34
B8. Additional Control Covariates . . . 36

A. Descriptive Statistics & Validation

A1. Summary Statistics

The estimation sample comprises 486 firm-call observations (286 pre-treatment, 200 post-treatment) across 65 firms (32 hardware, 33 digital). Table A1 reports mean tariff negativity by period, by sector and period, and by within-hardware upstreamness tercile; Table A2 lists the firms with the highest mean post-treatment tariff negativity.

Table A1: Descriptive Statistics: Tariff Negativity

Panel A. By period

Period	N	Mean	SD	Nonzero
Pre	286	0.0287	0.1170	21
Post	200	0.2080	0.2640	98

Panel B. By sector and period

Sector	N_{Pre}	N_{Post}	Pre	Post	Δ (Post−Pre)
Digital	143	100	0.0185	0.1250	0.1065
Hardware	143	100	0.0388	0.2910	0.2522
DiD (Hardware − Digital)					**0.1457**

Panel C. By upstreamness tercile (hardware firms)

Tercile	N_{Pre}	N_{Post}	Pre	Post	Δ (Post−Pre)
Low	46	35	0.0587	0.3870	0.3283
Mid	49	32	0.0214	0.2330	0.2116
High	48	33	0.0375	0.2460	0.2085

Notes: Unit of observation is the firm-call. "Nonzero" (Panel A) counts observations with strictly positive tariff negativity. Δ is the within-group post-minus-pre change in the mean; the DiD row in Panel B reports the raw (uncontrolled) 2×2 difference-in-differences. Panel C terciles are formed on fixed pre-treatment upstreamness within the 32 hardware firms; cell counts are small and the upstreamness gradient is estimated on continuous standardized upstreamness in the main regression (Model 3).

Table A2: Top 15 Firms by Post-Treatment Tariff Negativity

Rank	Firm	Tariff neg.	Upstreamness	Type
1	RTX	0.775	0.500	Hardware
2	QVCGA	0.700	−0.860	Digital
3	NVDA	0.662	0.520	Hardware
4	CNXC	0.633	0.067	Digital
5	AAPL	0.567	−0.600	Hardware
6	HPQ	0.550	−0.600	Hardware
7	AMAT	0.518	0.980	Hardware
8	EBAY	0.500	−0.720	Digital
9	GE	0.500	0.600	Hardware
10	SMCI	0.500	0.680	Hardware
11	MSI	0.467	0.075	Hardware
12	CDW	0.450	−0.200	Digital
13	ADI	0.438	0.750	Hardware
14	HWM	0.433	0.720	Hardware
15	AMZN	0.417	−0.120	Digital

Notes: Mean post-treatment tariff negativity by firm. Upstreamness is the fixed pre-treatment score.

A2. Sample Construction and Firm List

Our analysis uses 486 earnings call transcripts from 65 US-listed firms in the technology and defense sectors, covering nine consecutive quarters from Q1 2024 through Q1 2026. The sample comprises Fortune 500 firms in our target sectors with continuous quarterly reporting through the sample window. Transcripts were obtained from Capital IQ and parsed into a structured panel using custom Python preprocessing.

Firms are organized into the following sectoral tiers. The tier assignment is the primary input to the hardware indicator; the indicator itself is constructed at the firm level using the rule described below to handle within-tier heterogeneity.

- **Semiconductors (9):** ADI, AMD, AVGO, INTC, MCHP, MU, NVDA, QCOM, TXN
- **Semiconductor Equipment (3):** AMAT, KLAC, LRCX
- **Semiconductors/EMS (1):** SANM

- **Hardware/Systems (6):** AAPL, DELL, HPE, HPQ, SMCI, WDC
- **Networking/Cybersecurity (4):** APH, CSCO, MSI, PANW
- **Defense/Aerospace (10):** BA, GD, GE, HII, HWM, LHX, LMT, NOC, RTX, TDG
- **IT Services/Defense (5):** BAH, CACI, KBR, LDOS, SAIC
- **IT Services (7):** CDW, CNXC, CTSH, DXC, IBM, KD, NSIT
- **Software/Cloud (7):** ADBE, CRM, INTU, MSFT, NOW, ORCL, WDAY
- **Internet/E-Commerce (13):** ABNB, AMZN, BKNG, CHWY, CPNG, DASH, EBAY, EXPE, GOOGL, META, QVCGA, UBER, W

Hardware indicator construction. The hardware indicator equals 1 for firms whose primary revenue stream consists of physical goods that cross customs borders, and 0 otherwise. By tier, this includes all firms in Semiconductors, Semiconductor Equipment, Semiconductors/EMS, Hardware/Systems, and Defense/Aerospace. The Networking/Cybersecurity tier is split at the firm level by dominant product: Amphenol (APH), Cisco (CSCO), and Motorola Solutions (MSI) produce physical networking and electronic-component hardware and are coded as hardware; Palo Alto Networks (PANW) delivers primarily software and cloud-based cybersecurity and is coded as digital. The IT Services/Defense tier (BAH, CACI, KBR, LDOS, SAIC) is coded as digital, as these firms' revenue is predominantly from professional services and systems integration rather than physical goods crossing customs. The remaining digital tiers (IT Services, Software/-Cloud, Internet/E-Commerce) are coded as digital throughout. This procedure yields **32 hardware firms and 33 digital firms**.

The complete coded list:

- **Hardware (32):** AAPL, ADI, AMAT, AMD, APH, AVGO, BA, CSCO, DELL, GD, GE, HII, HPE, HPQ, HWM, INTC, KLAC, LHX, LMT, LRCX, MCHP, MSI, MU, NOC, NVDA, QCOM, RTX, SANM, SMCI, TDG, TXN, WDC

- **Digital (33):** ABNB, ADBE, AMZN, BAH, BKNG, CACI, CDW, CHWY, CNXC, CPNG, CRM, CTSH, DASH, DXC, EBAY, EXPE, GOOGL, IBM, INTU, KBR, KD, LDOS, META, MSFT, NOW, NSIT, NSIT, ORCL, PANW, QVCGA, SAIC, UBER, W

A3. Illustrative Tariff Discussion Excerpts

To validate the face validity of the LLM-coded tariff negativity measure, this section presents excerpts from earnings call transcripts with high LLM-assigned scores. The excerpts demonstrate that high-scoring observations correspond to substantive, interpretable discussion of tariff costs and policy uncertainty rather than superficial keyword matches.

NVIDIA (2025Q2): "On April 9, the U.S. government issued new export controls on H20, our data center GPU designed specifically for the China market.... We were unable to ship $2.5 billion in H20 revenue in the first quarter due to the new export controls.... Export controls should strengthen U.S. platforms, not drive half of the world's AI talent to rivals."

GE Aerospace (2025Q2): "The global aviation sector has long operated without tariffs on civil aircraft, engines and avionics. We'll continue to advocate for an approach that reestablishes zero-for-zero tariffs in the aviation sector.... The tariffs in aviation really aren't good for anybody."

Apple (2025Q3): "For the June quarter, we incurred approximately $800 million of tariff-related costs. For the September quarter, assuming the current global tariff rates, policies and applications do not change... we estimate the impact to add about $1.1 billion to our costs."

Raytheon (2025Q2): "The aerospace and defense sector has operated in a duty-free environment, and that has been instrumental to the industry maintaining one of the

largest trade surpluses across American manufacturing industries for decades.... With respect to China, at current U.S. and China tariff rates, we estimate there would also be a cost impact of around $250 million."

Boeing (2025Q2): "I would break this down into two categories: input tariffs that affect our cost to manufacture our products, and then the potential impact of retaliatory tariffs like those we are seeing in China.... The only region that we have an issue with aircraft delivery today is China, and due to the tariffs, many of our customers in China have indicated they will not take delivery."

HP Inc. (2025Q2): "Due to additional tariff costs that could not be fully mitigated in the quarter, our non-GAAP operating profit fell short of expectations.... Our operating margin was impacted by net tariff costs, mainly in Personal Systems."

Applied Materials (2025Q4): "The cost reductions were offset to some degree by tariff headwinds and by some of the inventory management."

A4. Illustrative Business Model Excerpts: Upstream vs. Downstream

To validate the face validity of the LLM-coded upstreamness measure, this section presents business model excerpts from firms at the upper and lower extremes of the upstreamness distribution. High-upstream firms describe themselves in terms of equipment, components, and manufacturing embedded within other firms' production processes. Low-upstream firms describe themselves in terms of users, guests, consumers, and customer experience.

Panel 1: High Upstream Firms (Inputs to Other Firms' Production)

Lam Research (0.99): "Our cobot technology offers an important and cost-effective solution to enable precise and repeatable maintenance beyond what human workers

alone can achieve... inside the fab."

Applied Materials (0.98): "Our innovations are enabling the major device architecture inflections that are critical to advancing energy-efficient AI... We track many of your end customers in the semiconductor industry, both fabless and IDMs."

KLA Corp. (0.96): "We have to deal with some inefficiency as we've staffed up to support those fabs and now we have to move those folks to support other customers... Logic/Foundry chip makers at the leading edge."

Amphenol (0.84): "This revolution in AI continues to create a unique opportunity for Amphenol given our leading high-speed and power interconnect products... manufacturing presence around the world, new factories in Southeast Asia, North America, Mexico, Eastern Europe, North Africa."

Micron (0.80): "Manufacturing elements that we're integrating together on the front end to be able to really deliver that higher performance... new technologies in process flow for assembly and test, which will enable us to deliver the high volumes of integrated HBM cubes."

Panel 2: Low Upstream / Downstream Firms (Final Products to End Users)

Apple (-0.60): "We love hearing from customers who are discovering for the first time the versatility of iPad, from the ultra-portable iPad mini built from the ground up for Apple Intelligence to the powerful M4 iPad Pro... iPad is there for our users whenever they need it."

Booking (-0.65): "At Booking Holdings, we have always been driven by innovation... consumer adoption of mobile apps to using sophisticated machine learning models early in our business, we have consistently evolved to meet the needs of travelers and partners."

Airbnb (-0.68): "We want to be very mindful of the guest journey and to be very

thoughtful with regard to both personalization and timing around what type of products are we merchandising to the customer."

eBay (−0.72): "As we look at our U.K. initiative. . . the changes we're making across key parts of the C2C experience will allow us to unlock the sizable market opportunity from the vast array of unique inventory sitting in U.K. consumers' homes."

Chewy (−0.85): "E-com has picked up a bigger share of that normalization and conversion relative to the other channels. . . there's a lot of innovation internally that is customer-facing."

The linguistic contrast between panels is consistent across the full sample: upstream firms describe themselves in terms of inputs, components, fabs, and integration into downstream production; downstream firms describe themselves in terms of users, guests, customer experience, and end-user satisfaction.

A5. Cross-Model Reliability: GPT vs. Claude

To assess robustness to LLM coder choice, we re-coded a random subsample of 50 transcripts ($N = 50$, ∼10% of the corpus) using Anthropic's Claude Sonnet 4.6, holding all prompts and scoring rubrics constant. The subsample was randomly drawn from the full corpus ($N = 486$) using a fixed seed for reproducibility, and both coders applied identical scoring prompts, response schemas, and temperature settings. Inter-coder reliability statistics are reported in Table A3.

We report two complementary measures of inter-coder agreement. Spearman's rank correlation (ρ) measures the strength of monotonic association between coders' scores, capturing whether the two models rank firms similarly regardless of the absolute magnitudes they assign. A Spearman ρ of 1.0 indicates that both coders produce identical orderings; values above 0.80 are conventionally interpreted as strong agreement (Schober, Boer, and Schwarte 2018). Pearson's r measures the strength of linear association and

Table A3: Inter-Coder Reliability: GPT-4o vs. Claude Sonnet 4.6

	Tariff Negativity	Upstreamness
Spearman ρ	0.961***	0.891***
Pearson r	0.978***	0.875***
Mean (GPT)	0.085	0.183
Mean (Claude)	0.052	0.010
SD (GPT)	0.191	0.606
SD (Claude)	0.119	0.628
N	50	50

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

is more sensitive than Spearman to differences in absolute scoring level. Reporting both statistics allows us to distinguish agreement on ordering (Spearman) from agreement on magnitudes (Pearson). Agreement on either measure exceeds conventional thresholds for inter-coder reliability with continuous measures (Krippendorff 2018).

For our research design, Spearman ρ is the more substantively relevant statistic. Our main specification uses within-firm and within-quarter variation, captured by firm and calendar-quarter fixed effects, which absorb any systematic level differences between coders. What matters for identification is whether the two LLMs produce consistent *rankings* of firms on the tariff negativity and upstreamness dimensions — not whether they produce identical numerical scores. The Spearman correlations of 0.961 for tariff negativity and 0.891 for upstreamness indicate that the rank ordering of firms is highly stable across coders. The corresponding Pearson correlations (0.978 and 0.875) further indicate that, conditional on rank ordering, the two coders also produce scores that are linearly proportional to one another. Together, these statistics provide strong evidence that our measures capture genuine structural and rhetorical differences across firms rather than model-specific artifacts of language interpretation.

The modest downward shift in Claude's mean scores, particularly for upstreamness, where the gap is approximately 0.17, reflects differences in how the two models interpret the scoring rubric's anchors, not disagreement about substantive content. Claude appears to apply a somewhat more conservative threshold for assigning extreme scores at both

ends of the scale, while GPT-4o extends more readily toward the boundaries. This produces a compression of Claude's distribution rather than reordering. Because our regression specification includes firm fixed effects, any constant difference in scoring level between coders is absorbed and does not affect the estimated treatment effects. The within-firm variation that drives our identification is preserved across coders, as evidenced by the high rank correlations.

We also examine agreement at the distribution extremes, where face validity considerations are sharpest. Among the ten firm-quarter observations with the highest GPT-coded upstreamness scores, nine receive Claude scores above 0.50, with Applied Materials (1.00), Lam Research (1.00), and Texas Instruments (0.80) achieving near-identical scores under both coders. The single exception is Dell, which scores 0.80 under GPT but 0.40 under Claude, reflecting a substantive ambiguity in classification, as Dell occupies an intermediate position selling both enterprise hardware (upstream-leaning) and consumer products (downstream-leaning). Among the ten firm-quarter observations with the lowest GPT-coded upstreamness scores, all ten receive Claude scores at or below 0.60, with consumer-facing platforms (Wayfair, Airbnb, Booking, Chewy, eBay) clustering at the bottom under both coders. This convergence at the distribution extremes, and particularly the strong agreement at the downstream end where consumer-facing firms cluster, is especially important for our identification strategy: the firms whose scores anchor the upstream and downstream ends of the distribution are precisely the firms whose contrast drives the within-hardware upstream effect documented in Model 3 of Table 1. These results provide strong evidence that our LLM coding procedure is robust to coder choice. While we use GPT as the primary coder due to its lower cost and faster throughput across the full 486-transcript corpus, the substantive findings would be expected to hold under Claude-coded measures, given the high rank correlation and preserved ordering at the distribution extremes.

A6. LLM Prompts

We reproduce the full set of prompts used in the LLM coding pipeline (in Python) below. The extraction prompt (Stage 1) is applied to the full transcript; the three scoring prompts (Stage 2) are applied to the extracted passages. We use the same prompt for both our baseline OpenAI and our cross-validiation Claude coding schemes.

Extraction Prompt

```
You are extracting relevant passages from an earnings call transcript.

TASK: Find and extract VERBATIM quotes for THREE categories, plus a company summary.

CATEGORY 1: TRADE & POLICY ENVIRONMENT (DIRECT)
Extract ANY passage that EXPLICITLY touches on trade, tariffs, or policy by name:
- Tariffs, duties, Section 301/232, trade war, trade policy
- Import/export restrictions, customs, trade tensions
- Reshoring, nearshoring, onshoring
- CHIPS Act, IRA, government subsidies tied to domestic production
- Export controls, sanctions, entity lists
- Trade agreements, negotiations, deals

CATEGORY 2: TRADE-ADJACENT LANGUAGE (INDIRECT/LATENT)
Extract passages that DO NOT name tariffs but discuss phenomena PLAUSIBLY
CONNECTED to the trade environment. Cast an EXTREMELY WIDE net:
- Input cost increases, component cost volatility, margin pressure from sourcing
- Supply chain restructuring, diversification, dual-sourcing, shifting production
- China business: revenue, customers, competition, operations
- Manufacturing geography: where products are made, why certain locations
- Geopolitical risk or uncertainty
- Government incentives, subsidies, grants, tax credits
- Competitive framing against foreign firms
- Inventory strategy changes plausibly linked to supply uncertainty
- Pricing actions attributed to cost environment
```

```
- Capital expenditure shifts toward domestic capacity
- ANY mention of specific countries in supply chain context

CATEGORY 3: BUSINESS MODEL
- What products/services the company makes
- Who their customers are
- Revenue breakdown by segment/customer type
- Position in supply chain

COMPANY DESCRIPTION: Always provide a one-sentence description.

RULES:
- Extract VERBATIM quotes, 1-3 sentences each, max 100 words
- Up to 10 quotes per Category 1, up to 15 per Category 2
- When in doubt, INCLUDE IT
- ALWAYS provide company_description
```

Tariff Negativity Prompt

```
You are measuring how NEGATIVELY a firm frames tariffs and trade restrictions.

TARIFF_NEGATIVITY (0-1):
0.0 = no mention or positive/neutral framing
0.1-0.2 = passing mention with mild concern
0.3-0.4 = clear discussion of costs/challenges
0.5-0.6 = meaningful business problem, specific costs named
0.7-0.8 = serious threat, financial impact quantified
0.9-1.0 = severely damaging, explicit opposition

SILENCE = 0.0. Score OBSERVABLE TEXT only.

TARIFF_IMPACT (-1 to +1): Economic effect regardless of framing.
+1=strong beneficiary, -1=severely harmed, 0=neutral.
```

```
TARIFF_SALIENCE (0-1): How prominently tariffs feature.
0=not mentioned, 1=dominant topic.

EVIDENCE_TYPE: explicit | implied | contextual | no_signal
Use fine-grained scores.
```

Upstreamness Prompt

```
Determine firm's position in the PRODUCTION VALUE CHAIN.

UPSTREAM (+1): Makes INPUTS for other companies' products
  (chips, components, cloud infrastructure, enterprise software platforms)
DOWNSTREAM (-1): Makes FINAL PRODUCTS for end users
  (consumer electronics, consumer software, retail, streaming)
MIDSTREAM (0): Mixed

SCORING:
+1.0 = Pure upstream (TSMC)
+0.8 = Heavily upstream (Qualcomm)
+0.6 = Mostly upstream (Intel)
+0.4 = Somewhat upstream (Nvidia)
+0.2 = Slight upstream lean (Microsoft)
 0.0 = Mixed
-0.2 = Slight downstream lean (Amazon)
-0.4 = Somewhat downstream (Meta)
-0.6 = Mostly downstream (Apple)
-0.8 = Heavily downstream (Netflix)
-1.0 = Pure downstream (Snap)

B2B  upstream. Key question: do products become INPUTS in other products?
CUSTOMER_TYPE: other_businesses | consumers | both | government
You MUST provide a score.
```

Latent Trade Environment Prompt

```
You are analyzing earnings call excerpts to measure how the TRADE POLICY
ENVIRONMENT manifests in a firm's communication. This is BROADER than
explicit tariff attitudes.

Score SEVEN dimensions:

1. TRADE_ENV_SALIENCE (0-1): How prominently does the firm discuss
trade-environment-adjacent topics? 0=none, 1=dominant theme.
Score based on PRESENCE, not valence.
0.0 = no trade-relevant discussion
0.05-0.15 = one or two brief mentions
0.2-0.3 = multiple references, some discussion
0.4-0.5 = clearly important topic
0.6-0.7 = major presence, shapes narrative
0.8-1.0 = dominant theme of the call

2. TRADE_NEGATIVITY_VALENCE (-1 to +1): Tone of trade-environment language.
-1=entirely negative, 0=neutral, +1=entirely positive.
If salience=0, set to 0.

3. SUPPLY_CHAIN_SHIFT (0-1): Discussion of restructuring/relocating supply chain.
4. CHINA_DISCUSSION (0-1): ANY discussion of China business/risk/opportunity.
5. COST_ATTRIBUTION (0-1): Attribution of costs to external/macro/policy factors.
6. MANUFACTURING_LOCATION (0-1): Discussion of WHERE firm manufactures and why.
7. GOV_INCENTIVE_DISCUSSION (0-1): Discussion of government subsidies/incentives.

EVIDENCE_TYPE: explicit_trade | trade_adjacent | contextual | no_signal
CONFIDENCE: 0.8+ extensive, 0.5-0.8 moderate, 0.3-0.5 limited, 0.1-0.3 weak

Use fine-grained scores (0.12, 0.35, 0.57). Score ALL dimensions.
```

The prompt for extracting upstreamness information from a different source (10-K

filings, and the rationale is presented in the section below) also shares the same skeleton, with adjustment to improve the precision, given the different nature of 10-K filings and earnings call. The model was GPT-4o, and the supplied text consisted only of the firm's extracted Form 10-K Item 1 (Business) section. Note that unlike the main prompt for our manuscript, the original production prompt used for the 10-K validation is very long. For readability and space, we present a condensed version that is substantially shorter than the original one yet still preserving the full substantive coding logic, while omitting repetitive instructions and implementation-oriented language.

10-K Upstreamness Validation Prompt

```
You are analyzing a firm's 10-K Item 1 (Business) section to determine its
STRUCTURAL POSITION in the PRODUCTION VALUE CHAIN.

DEFINITION OF UPSTREAMNESS:
Where do the firm's economically important outputs sit in the chain from
raw materials -> intermediate inputs -> productive infrastructure ->
final goods/services -> final users?

UPSTREAM (+1): Primarily supplies intermediate inputs, components,
production equipment, infrastructure, or productive capabilities used by
other organizations to produce or deliver their own goods/services.

DOWNSTREAM (-1): Primarily supplies final goods/services whose economic
purpose is fulfilled through use or consumption by the purchaser/end user.

MIDSTREAM (0): Genuinely mixed or structurally intermediate.

CORE RULE: Do NOT ask only "Who buys the product?" Ask what happens
ECONOMICALLY to the output after it is sold. Does the customer obtain an
INPUT/PRODUCTIVE CAPABILITY used to produce or deliver something else, or
the FINAL GOOD/SERVICE it intends to use?
```

OUTPUT CLASSIFICATION:

1. Intermediate components/material inputs
   (chips, memory, sensors, connectors) -> strongly upstream.
2. Contract manufacturing/production capacity -> strongly upstream.
3. Specialized production equipment
   (semiconductor fabrication/process-control equipment) -> strongly upstream.
4. Productive infrastructure
   (servers, networking, storage, data centers, cloud infrastructure)
   -> upstream when it enables customers' production/delivery.
5. Enterprise software/business services -> upstream only when they
   materially enable production/delivery; administrative/final-use services
   should be closer to 0.
6. Complete operational systems
   (aircraft, warships, military platforms, corporate PCs) -> determine
   whether they function as productive capability or terminal organizational
   use; do NOT automatically classify them as upstream.
7. Final consumer goods/services -> downstream.

IMPORTANT:

B2B != upstream.

B2G != upstream.

Physical incorporation is NOT required for upstreamness.

Technical sophistication != upstream.

Capital intensity != upstream.

Defense production != upstream.

ZERO IS NOT UNCERTAINTY:

Use 0.0 only when the firm's structural position is genuinely balanced or
intermediate. If evidence points upstream/downstream but is incomplete,
preserve the directional score and reduce CONFIDENCE instead.

DEFENSE/GOVERNMENT FIRMS:

Distinguish components, subsystems, propulsion, sensors, electronics,
production equipment, mission infrastructure, integration, and complete

systems. Components/subsystems receive positive upstream weight; complete terminal systems receive less. Weight the firm's overall mix rather than classifying defense firms automatically as upstream or neutral.

DIVERSIFIED FIRMS:
Assess the WHOLE FIRM. Identify major segments, classify each segment's economic role, and weight by revenue or business importance when available. Do not let one vivid product or small segment determine the firm-level score.

SCORING:
+1.0 = Almost entirely intermediate inputs/production equipment (TSMC/AMAT)
+0.8 = Heavily upstream (Qualcomm)
+0.6 = Mostly upstream (Intel)
+0.4 = Meaningfully upstream/productive infrastructure (Nvidia)
+0.2 = Slight upstream lean (Microsoft)
 0.0 = Genuinely mixed or structurally intermediate
-0.2 = Slight downstream lean (Amazon)
-0.4 = Somewhat downstream (Meta)
-0.6 = Mostly downstream (Apple)
-0.8 = Heavily downstream (Netflix)
-1.0 = Almost entirely final-user goods/services

Use the full continuous range from -1.0 to +1.0; scores need not be in increments of 0.2. Calibration examples are scale anchors only.

Base the classification ONLY on the supplied 10-K text. Do not use outside knowledge about the company.

CUSTOMER_TYPE: other_businesses | consumers | both | government
EVIDENCE_TYPE: explicit | inferred | weak
CONFIDENCE: 0.8-1.0 direct/extensive, 0.5-0.8 reasonably clear, 0.3-0.5 substantial inference, 0.1-0.3 weak/conflicting

You MUST provide a score.

10-K Import Dependence Prompt

```
You are analyzing passages from a firm's annual 10-K filing that describe
its supply chain, manufacturing, suppliers, and facilities. Your task is
to measure the firm's FOREIGN INPUT DEPENDENCE: how much the production
of what the firm sells (or the infrastructure it runs on) depends on
manufacturing, assembly, components, or raw materials located OUTSIDE
the United States.

This is a STRUCTURAL measure of the supply chain. Do NOT score the firm's
tone, worry, or commentary about tariffs or trade policy. Two firms with
identical supply chains must receive the same score even if only one of
them discusses tariffs.

WHAT COUNTS AS FOREIGN INPUTS:
- Finished products manufactured or assembled abroad (in-house or by
  contract manufacturers/ODMs)
- Chips or wafers fabricated by foreign foundries
- Components, sub-assemblies, or raw materials sourced from foreign
  suppliers
- For service/software/platform firms: imported hardware their
  operations depend on (servers, networking equipment, devices), if the
  filing indicates material reliance on it

SCORING GUIDE:
  1.0 = Core products are almost entirely manufactured or assembled
        abroad and imported; key components come from a few foreign
        suppliers.
  0.8 = Core products are made by foreign foundries or contract
        manufacturers (e.g., a fabless design model, outsourced overseas
        assembly), with heavy reliance on overseas supply.
  0.6 = Substantial foreign production or foreign components alongside
        meaningful domestic production.
  0.4 = Production is mostly domestic, but with material reliance on
```

```
        imported components or raw materials; OR a services/platform firm
        whose core infrastructure relies heavily on purchased hardware
        that is largely produced abroad.
  0.2 = Mostly domestic production or services; imported inputs are
        limited or secondary.
  0.0 = No physical production and negligible dependence on imported
        inputs.

GUIDANCE:
- A firm with large foreign facilities that mainly serve foreign markets
  is less dependent on imports INTO the United States than one that
  produces abroad for the U.S. market. Weigh this when the filing makes
  it clear.
- "Global supply chain" boilerplate alone is weak evidence. Prefer
  specific statements about where and by whom products are made.
- Use intermediate values (e.g., 0.3, 0.7) when the firm sits between
  anchors.

CONFIDENCE: 0.8+ explicit statements, 0.5-0.8 clear but partial,
0.3-0.5 inferred, 0.1-0.3 weak.

You MUST provide a score.
```

A7. Human Validation

To ensure that our LLM-coded DV is precise, we get an research assistant to classify the tariff negativity based on earnings call scripts. Specifically, our research assistant was assigned 50 randomly selected transcripts from our sample (similar to the number of observations in our cross-model check, for consistency). Then the assistant was given a guidance on the rubric to score the negativity in the scale of 0 to 1, similar to the LLM prompt. To ensure that the variable is free-of-bias and independently validated, the assistant did not have any information on the LLM-coded values, and she has to justify

her scoring. We use the same metrics as our cross-model validation above, and compare the human-coded negativity with the GPT-coded value. Our results are presented in the Table A4 below.

Table A4: Human Validation of LLM-Coded Tariff Negativity

	Spearman ρ	Pearson r
Tariff negativity (overall)	0.64^{***}	0.45^{***}
Tariff presence (exact agreement)	94%	

Notes: "Tariff presence" is exact agreement on whether a transcript contains any negative tariff discussion.

Overall, the correlation between human and GPT-coded value is acceptable, though weaker than that in the cross-model test. Human and model agree strongly on transcripts that actually discussed tariffs (94%)). The agreement with the ordering of negativity is relatively decent with 0.64 of Spearman score, meaning the assistant and the LLM are reasonably in line with each other on how they rank firms' tariff negativity. The Pearson score, which is about the linearity between two sets of raw score, while positively significant, is weaker than the Spearman one. This weaker linear relationship means that there is some degree of disagreement between the research assistant and the LLM. Although a more moderate relationship is far from ideal in our validation, this does not pose a significant problem to our key results. As tariff negativity is originally a continuous, fine-grained score, a low Pearson score is expected: human could catch the nuance and language subtlety very well, but the exact scoring on a continuous scale could be subjective and less standardized than the LLM, which leads to some discrepancies in the coding. Because the sample size is small, the results are highly sensitive to outliers, which, combined with the human subjectivity, explains the moderately weak Pearson score. Additionally, the Pearson score is still positively significant, and being juxtaposed with the Spearman score, we could still safely conclude that the human assistant and our LLM is still relatively in agreement with each other, which validates our LLM-coded values.

A8. Upstreamness Measurement Validation

To avoid same-source bias given that our outcome and explanatory variables come from the same transcripts, we adopt the same LLMs approach to code the firm-level upstreamness from a different source: 10-K filings. The 10-K form is a detailed annual report that American public companies must file to the U.S. Securities and Exchange Commission (SEC) to share their financial health and business operations. It tells us what the business does, lists its physical properties, and notes any major lawsuits, as well as provides information about its financial statements, market, and management's view on past results as well as other issues related to the business. We therefore use the same GPT model that we use for our main analysis to extract the upstreamness of a firm from the Item 1 of its 10-K, during the same time period. Besides providing the Pearson and Spearman scores for both sources, we also re-run the whole analysis on this newly-coded upstreamness, and the results are below.

Table A5: Cross-Source Validation of Upstreamness

	Earnings Calls	10-K Item 1
Spearman ρ	0.916^{***}	
Pearson r	0.919^{***}	
Mean	0.214	0.186
SD	0.531	0.523
N	65	65

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Similar to our previous validations, both the Spearman and Pearson scores in this test demonstrate a high level of significance and magnitude, meaning that these independent sources have strong convergence. Furthermore, slight differences in their mean (approximately 0.03) and their standard deviation (0.008) also confirm that the coded upstreamness is not the artifact of sources. The same-specification results are also mostly identical to our main models, which further strengthens our construct validity. Additionally, we provide another same-specification set of results that exclude firms that have a large disagreement across two sources (>0.3), and our findings are still robust, meaning that they are not driven by some extreme values coded from another source.

B. Robustness Tests

B1. Binary Dependent Variable

We binarize our dependent variable, 1 if a firm has a negative attitute towards tariffs in a given transcript (tariffs_negativity > 0), and 0 otherwise. This is the *extensive margins* of how salient tariffs are to their business, albeit less precise than our baseline outcome, which emphasizes the *intensive margins*. We re-run all of our specifications in the main text with this new variable, and the results are below.

Overall, the results are well aligned with our main specifications, with one small caveat. The effect of upstreamness within hardware firms after Liberation Day, while still negative, are less significant than our baseline (only at 90 %). However, we contend that this is far from weakening our key findings. The drop in significance of upstreamness within hardware firms can be attributed to the coarser coding scheme for the outcome variable, because collapsing tariff negativity to an indicator discards the within-positive variation. Under this new operationalization, two hardware firms that differ in the intensity of their tariff negativity, which is what the variation H2a captures, are coded identically once the measure is binarized. The binary DV therefore tests the extensive margin (whether any negativity is expressed), on which the gradient is necessarily weaker,

Table B1: Robustness: Binary Dependent Variable

	(1) H1	(2) H2	(3) H2 + H2a
Upstream × Post	0.024		
	(0.043)		
Hardware × Post		0.237***	0.281***
		(0.081)	(0.079)
Upstream × Hardware × Post			−0.079*
			(0.046)
Firm FEs	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes
Observations	486	486	486

Notes: Standard errors clustered at the firm level in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

while the level effect, which also operates on the extensive margin, is fully preserved.

B2. Exclusion of Defense Firms

To verify that our results are not an artifact of the defense sector's distinctive relationship to trade policy, we re-estimate our specifications excluding all defense-related firms, both aerospace and military-hardware primes (Defense/Aerospace) and defense service contractors (IT Services/Defense). These two groups have tariff posture which is governed by forces largely orthogonal to the customs-exposure mechanism we identify. Aerospace and equipment primes operate in a near-monopsonic environment in which the U.S. government is the dominant customer; this relationship shapes their public stance on trade policy through fixed-price procurement and domestic-sourcing mandates such as the Buy American Act and specialty-metals provisions - channels unrelated to global value chain position (Sandler and Hartley 1995; Rogerson 1994; Moran 1990; Congressional Research Service 2024). Their tariff attitudes vary widely as a result of the commercial-versus-government distinction: firms with large civilian-aerospace businesses (Raytheon, General Electric) respond sharply to tariffs, while pure-play government primes remain

near-silent[1]. Defense service contractors, by contrast, earn revenue from professional services rather than goods crossing customs, leaving them with negligible tariff exposure and a flatter distribution of tariff negativity[2].

Table B2: Robustness: Difference-in-Differences Estimates Excluding Defense-Related Firms

	(1) Baseline	(2) H2	(3) H2 + H2a
Upstream × Post	−0.003		
	(0.029)		
Hardware × Post		0.122**	0.197***
		(0.052)	(0.048)
Upstream × Hardware × Post			−0.110***
			(0.027)
Firm FEs	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes
Observations	379	379	379
Firms	50	50	50

Notes: Sample excludes all defense-related firms (Defense/Aerospace and IT Services/Defense tiers), retaining 50 firms (22 hardware, 28 digital). Upstreamness is re-standardized within the restricted sample. Standard errors clustered at the firm level in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The results, reported in Table B2, are unchanged in structure. The pooled baseline remains a precise null, while the hardware level effect and the within-hardware upstreamness effect are both significant and identical in sign to our main estimates. The pattern across the three specifications mirrors the full sample exactly: no pooled relationship between upstreamness and tariff negativity, a sharp hardware-versus-digital divergence, and an attenuating effect of upstream position within hardware firms. Notably, this holds on a substantially smaller sample — 50 firms rather than 65 — which only sharpens the inferential demands, yet the estimates remain well-identified. Because the two excluded tiers fall on opposite sides of the hardware/digital divide and are shaped by mechanisms distinct from the customs-exposure channel we propose, the persistence of these effects

1. Within Defense/Aerospace, firm-level upstreamness and post-treatment tariff negativity are uncorrelated ($r = -0.07$, $n = 10$), and within the tier there is a high dispersion (SD = 0.24).

2. The two defense-related tiers occupy opposite extremes of within-tier dispersion in post-treatment tariff negativity: Defense/Aerospace exhibits the highest variation in the sample (SD = 0.24), while the services-based IT Services/Defense tier is tightly clustered near zero (SD = 0.08).

indicates that they are driven by the commercial hardware firms central to our argument rather than by the defense sector's idiosyncratic political economy.

B3. Placebo Test

We conduct two placebo tests to confirm that our estimates capture the effect of the Liberation Day tariffs rather than pre-existing differences between hardware and digital firms. We reassign the treatment to two earlier cutoffs, Q2 2024 and Q4 2024, and re-estimate Model 3 on the pre-treatment subsample. The two dates deal with different concerns. Q2 2024 was when the trajectory of trade policy remained uncertain so a genuine Liberation Day effect should leave no trace at this date. Q4 2024 is more demanding: by then Trump had won the election, and given the centrality of tariffs to his agenda, firms may have begun anticipating and responding to the coming policy before it took effect. A significant placebo effect at Q4 2024 would suggest that the hardware-digital divergence we attribute to Liberation Day in fact began with electoral anticipation. A null result, by contrast, indicates that any anticipation was not yet differential across firm types, and that the divergence is specific to the policy shock itself.

Table B3: Placebo Tests: Fake Treatment Dates in the Pre-Treatment Period

	(1) Real Treatment (Q2 2025)	(2) Placebo (Q2 2024)	(3) Placebo (Q4 2024)
Hardware × Post	0.184^{***}	-0.006	0.033
	(0.042)	(0.022)	(0.036)
Upstream × Hardware × Post	-0.082^{***}	-0.008	-0.017
	(0.025)	(0.022)	(0.032)
Firm FEs	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes
Sample	Full	Pre-only	Pre-only
Observations	486	286	286

Notes: Column (1) reproduces our main estimate in the main text. Columns (2) and (3) replace the treatment with fake dates in the pre-treatment period (Q2 2024 and Q4 2024 respectively), estimated on the pre-treatment subsample only. Standard errors clustered at the firm level in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

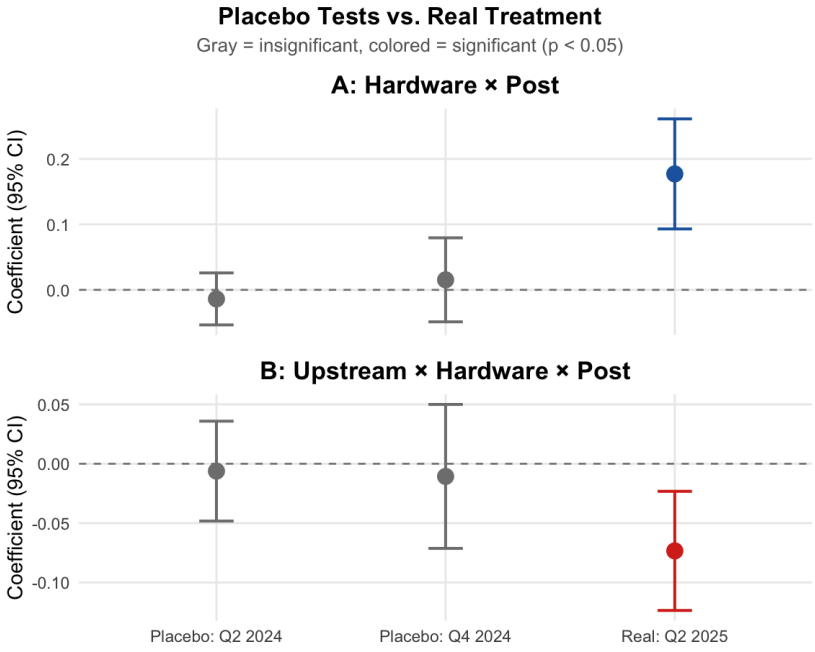


Figure B1: Placebo and Real Effects

Table B3 and Figure B1 report the results alongside the real estimate. At both placebo dates - Q2 2024 and Q4 2024 - neither the hardware level effect nor the within-hardware upstreamness effect is distinguishable from zero: the point estimates cluster tightly around the dashed zero line, and every confidence interval comfortably contains it. The contrast with the actual Liberation Day estimate is stark. The real hardware effect (Panel A) sits an order of magnitude above its placebo counterparts, and the real upstreamness effect (Panel B) is the only estimate whose confidence interval excludes zero, falling well below the placebo intervals that straddle it. The placebo treatments also explain almost none of the within-firm variation in tariff negativity. Together with the joint pre-trends test, these results indicate that the divergence in tariff attitudes is specific to the Liberation Day policy shock and not an artifact of trends already underway before treatment.

B4. Two-way Clustered Standard Errors: Firm and Calendar-Quarter

Our main specifications cluster standard errors at the firm level, allowing for arbitrary within-firm correlation across quarters. This may understate uncertainty if tariff negativity is also correlated across firms within a given quarter. For instance, if a macroeconomic announcement or a common news cycle moves many firms' earnings-call sentiment simultaneously. To tackle this issue, we cluster on both firm and calendar quarter, allowing correlation within firms over time and within quarters across firms. We note that the quarter dimension comprises only nine clusters, well below the threshold at which cluster-robust inference is reliable on its own. As such, we treat this specification as complementary to the firm-level small-sample corrections reported above (CR2 standard errors and the wild cluster bootstrap) rather than as a substitute for them. We expect that the results would align well with our main specification.

Table B4: Robustness: Two-Way Clustered Standard Errors (Firm and Quarter)

	(1) Firm-Clustered	(2) Two-Way Clustered
Hardware × Post	0.184^{***}	0.184^{*}
	(0.042)	(0.062)
Upstream × Hardware × Post	-0.082^{***}	-0.082^{**}
	(0.025)	(0.020)
Firm FEs	Yes	Yes
Quarter FEs	Yes	Yes
Clustering	Firm	Firm & Quarter
Observations	486	486

Notes: Column (1) clusters standard errors by firm (65 clusters); column (2) clusters by both firm and calendar quarter (65 and 9 clusters respectively). Standard errors in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table B4 reports the result. Allowing for within-quarter correlation across firms leaves both estimates significant: the hardware level effect remains the same, so does the within-hardware upstreamness effect. The point estimates are unchanged, as clustering affects only the standard errors, and the standard error on the upstreamness effect in fact

declines slightly under two-way clustering, indicating that within-quarter correlation is modest and contributes little additional uncertainty. The persistence of both effects under this less conservative error structure suggests that the results are not driven by common shocks affecting many firms in the same quarter, but reflect the genuinely differentiated response of hardware firms to the tariff regime.

B5. Alternative DV: Tariff-Related Lobbying

Our main analysis rests on a difference-in-differences design in which both the dependent variable (tariff negativity) and a key independent variable (upstreamness) are coded by a large language model from earnings-call transcripts. This approach raises two related concerns. First, both the outcome and the regressor originate from the same text-analysis pipeline, so any systematic measurement error in the model's coding, a tendency to associate certain rhetorical styles with both negativity and value-chain position, for instance, could induce a spurious correlation that the design would not detect. Second, and more broadly, expressed attitudes in earnings calls are a form of communication; a skeptic might argue that what we capture is strategic talk directed at investors rather than firms' underlying preferences, raising questions about whether our findings extend to consequential political behavior. A robustness analysis that replicates our results using an outcome that is both unambiguously behavioral and drawn from a independent data source of the LLM pipeline would address both concerns directly.

We construct a firm-quarter panel from Senate Lobbying Disclosure Act filings (57 firms, 513 firm-quarters, 2024Q1–2026Q1), where the outcome variable is the share of each firm's quarterly lobbying activities that mention tariffs, and we mirror our main specification with firm and calendar-quarter fixed effects and firm-clustered standard errors. Lobbying is a costly, legally disclosed action rather than cheap talk, and it is recorded in administrative filings that owe nothing to our text pipeline. If the same pattern appears in lobbying behavior, the concern that our main results are an artifact of LLM-based measurement is largely mitigated.

Estimating this design requires attention to treatment timing, because lobbying and expressed preferences respond to different moments. Our main analysis dates the treatment to Liberation Day (2025Q2), when the tariff regime was actually imposed. For lobbying, however, the relevant shock can arrive a quarter earlier, in 2025Q1, for two reasons. First, firms routinely intensify lobbying at the onset of a new administration in order to reestablish access to incoming officials; because this cannot occur until appointees are in place, it necessarily begins in the first quarter of the new term. Second, the incoming administration had signaled throughout the campaign that aggressive trade policy would be a centerpiece of its agenda, giving exposed firms clear incentive to lobby *before* the policy was finalized, while its design remained open to influence. The same logic explains why the two outcomes respond at different times. In 2025Q1 the eventual policy remained uncertain and, by historical precedent, was expected to prove less radical than campaign rhetoric implied; it would therefore have been premature for firms to voice dire assessments to investors in their earnings calls. Lobbying, which is forward-looking and aimed at shaping policy before enactment, responds to the *anticipated* shock; expressed preferences, which report realized conditions to investors, respond to the *actual* shock in 2025Q2, once Liberation Day made the consequences concrete. We therefore present both timings: Table B5 dates treatment to Liberation Day (2025Q2), matching our main analysis, and Table B6 dates it to the administration change (2025Q1).

Under both timings, the estimates are directionally consistent with our main results: hardware firms increase tariff lobbying relative to digital firms, and the within-hardware upstream interaction is negative, echoing the rhetorical forbearance of upstream firms. Under the Liberation Day timing (Table B5), the hardware effect is positive and significant and the triple interaction is negative but at threshold. Re-dating treatment to the administration change (Table B6) sharpens both estimates and improves model fit. That the pattern recovered from administrative lobbying records mirrors the pattern recovered from LLM-coded earnings calls is reassuring since the two designs share neither a dependent variable nor a data source, yet they point to the same structural divide.

The two timings differ in whether they satisfy the parallel-trends assumption, and in instructive ways. Under the Liberation Day timing, an event-study check reveals pre-treatment movement: the hardware leads are jointly marginally significant (Wald $p = 0.06$), reflecting that lobbying had already begun to rise in 2025Q1, before the formal imposition of tariffs. This is the anticipatory effect described above. Re-dating treatment to 2025Q1 absorbs this anticipation: the 2024 pre-treatment leads for the hardware interaction are then jointly indistinguishable from zero (Wald $p = 0.46$), and the hardware differential jumps sharply at 2025Q1 and remains elevated thereafter. For the hardware-digital distinction, then, the re-timed design satisfies parallel trends and supports a causal interpretation, in data wholly independent of our earnings-call measure.

We are more cautious about the within-hardware upstreamness interaction. Even under the re-timed specification, a joint test of its 2024 pre-treatment leads is marginally significant (Wald $p = 0.05$), indicating that parallel trends do not cleanly hold for this finer comparison. We therefore read the upstreamness component of the lobbying analysis as suggestive rather than causally identified, consistent with our broader finding that upstreamness operates principally on firms' expressed preferences and on their sorting across policy instruments, rather than on the dynamics of tariff-lobbying volume. To recapitulate, the lobbying analysis provides independent behavioral story of the divide between hardware and software for our argument: the result is robust to a shift from an LLM-coded attitudinal outcome to a disclosed behavioral one, drawn from an entirely separate data source. The upstreamness mechanism, by contrast, rests primarily on the expressed-preference evidence of our main analysis, where it is most cleanly identified.

Table B5: Lobbying Difference-in-Differences, Treatment at Liberation Day (2025Q2)

	(1) Baseline	(2) Hardware	(3) Triple
Upstream × Post	−0.007		
	(0.008)		
Hardware × Post		0.019	0.031**
		(0.012)	(0.013)
Upstream × Hardware × Post			−0.026*
			(0.014)
Firm FEs	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes
Observations	513	513	513

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B6: Lobbying Difference-in-Differences, Treatment at Administration Change (2025Q1)

	Tariff Lobbying Share
Hardware × Post	0.040**
	(0.017)
Upstream × Hardware × Post	−0.034*
	(0.020)
Firm FEs	Yes
Quarter FEs	Yes
Observations	513

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

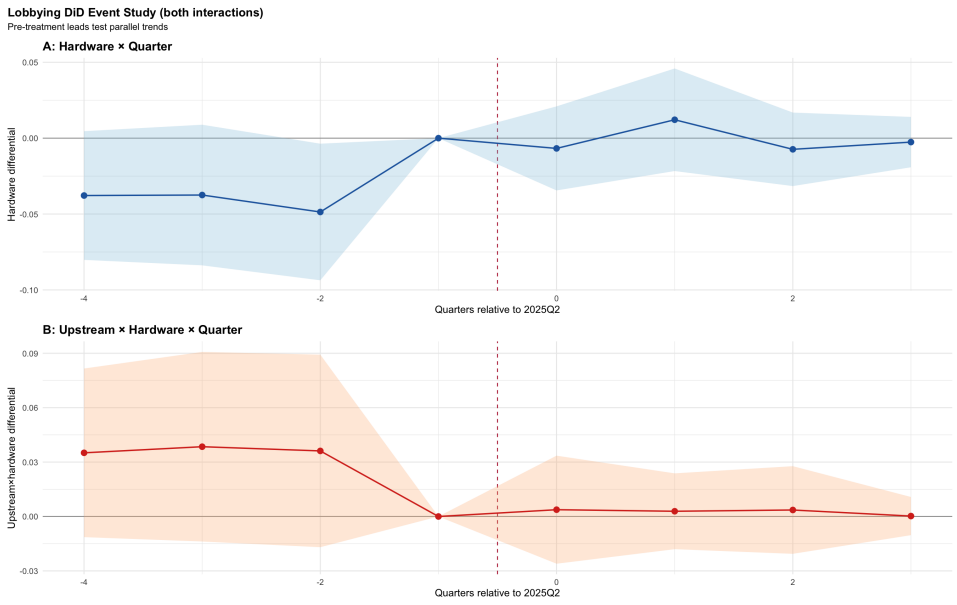


Figure B2: Event Study (Original Treatment)

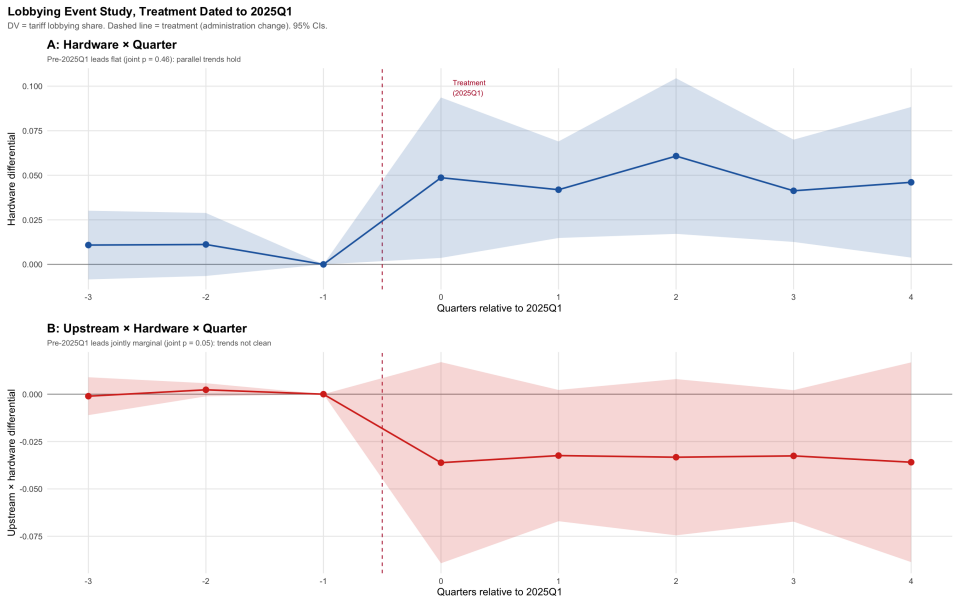


Figure B3: Event Study (Q1 2025 as a Treatment)

B6. Wild-Cluster Bootstrap, CR2 Standard Errors and Leave-One-Out Estimate

Table B7: Wild Cluster Bootstrap Inference for the Within-Hardware Interaction

	Upstream × Hardware × Post
Point estimate	−0.082
Cluster-robust t-statistic	−3.02
Bootstrap p-value	0.031
Reject H_0 at 5%	Yes
Bootstrap replications	9,999
Weights	Webb 6-point
Clusters (firms)	65

Notes: Wild cluster bootstrap-t for the triple-interaction coefficient from the preferred specification, imposing the null and resampling residuals at the firm level with Webb (2023) six-point weights over 9999 replications. The procedure provides finite-sample-valid inference with a moderate number of clusters, under which asymptotic cluster-robust standard errors can be unreliable.

Table B8: CR2 Bias-Corrected Cluster-Robust Inference

	Hardware × Post	Upstream × Hardware × Post
Point estimate	0.184	−0.082
CR2 standard error	0.060	0.028
t-statistic	3.07	−2.97
Satterthwaite d.f.	11.18	4.64
p-value	0.011	0.034
Significant at 5%	Yes	Yes

Notes: Bias-corrected cluster-robust (CR2) standard errors with Satterthwaite degrees-of-freedom corrections, clustered by firm (Bell and McCaffrey 2002; Pustejovsky and Tipton 2018). CR2 adjusts for the downward bias of conventional cluster-robust standard errors under a moderate number of clusters, and the Satterthwaite correction yields effective degrees of freedom reflecting the concentration of identifying variation. The low effective degrees of freedom for the triple interaction (4.64) indicate that it is identified from a limited number of high-leverage firms at the extremes of the value-chain distribution; the coefficient nonetheless remains significant at the 5% level. Estimates are from the preferred specification with firm and calendar-quarter fixed effects.

Table B9: Leave-One-Firm-Out Estimates of the Within-Hardware Interaction

	Upstream × Hardware × Post
Baseline estimate (all firms)	−0.082
Across 32 leave-one-firm-out fits:	
Estimate range	[−0.097, −0.064]
p-value range	[0.0005, 0.019]
Fits significant at 5%	32 of 32
Fits significant at 10%	32 of 32
Most attenuating exclusions (smallest $\|\hat{\beta}\|$):	
Drop Apple (AAPL)	−0.064 ($p = 0.009$)
Drop Micron (MU)	−0.076 ($p = 0.003$)
Drop Motorola Solutions (MSI)	−0.077 ($p = 0.004$)
Drop HP (HPQ)	−0.077 ($p = 0.019$)
Drop Broadcom (AVGO)	−0.078 ($p = 0.003$)

Notes: Each row drops one hardware firm and re-estimates the preferred specification. The coefficient remains negative and significant at the 5% level in all 32 fits. The exclusions that most attenuate the estimate are firms at the endpoints of the upstreamness distribution: downstream producers (Apple, HP) and upstream suppliers (Micron, Broadcom). This is consistent with an effect identified across the full range of value-chain positions rather than driven by any single firm. Firm and calendar-quarter fixed effects included throughout; standard errors clustered by firm.

B7. Specification with 10-K Upstreamness

Here we provide same specification as in our main text, with the independent variable coded from the 10-K filings. While the Spearman and Pearson scores confirm the construct validity, it is necessary to ensure that the variable works well in the actual analysis, as these variables are coded from different sources. Table B10 present results with a full sample. Because there is some large disagreement for some firms, we will run another analysis with these firms dropped out (only 7 out of 65) to see whether the results are driven by variance. Table B11 present results for refined sample. Both remain statisti-

cally significant, and the coefficients' magnitude is barely different from our main model. This robustness solidifies our upstreamness classification.

Table B10: 10-K Upstreamness Robustness

	Tariff Negativity
Hardware × Post	0.188***
	(0.052)
Upstreamness × Hardware × Post	−0.0842**
	(0.0389)
Firm FE	Yes
Quarter FE	Yes
Observations	486
Firms	65

Standard errors clustered by firm in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table B11: Refined 10-K Results in the Agreement Sample

	Tariff Negativity
Hardware × Post	0.1952***
	(0.0482)
Upstreamness × Hardware × Post	−0.0966***
	(0.0324)
Firm FE	Yes
Quarter FE	Yes
Observations	432
Firms	58

Notes: The 10-K upstreamness measure is re-standardized within this restricted sample. All specifications include firm and calendar-quarter fixed effects. Standard errors, in parentheses, are clustered by firm. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

B8. Additional Control Covariates

In our parsimonious model, firm and time fixed effects absorb time-invariant firm characteristics and common temporal shocks. One might still worry that firms differing in size, capital intensity, or dependence on foreign inputs trended differently after the shock for reasons unrelated to their value-chain position. We address this by interacting pre-treatment firm characteristics with the post indicator. Size is measured by log revenue and log assets, and capital intensity by capital expenditure relative to revenue, all for fiscal year 2024 and retrieved from firms' SEC 10-K filings.

The last of these concerns matters most for our argument: upstream hardware firms might be less negative about tariffs simply because they depend less on imported inputs. To address it, we measure each firm's foreign input dependence from the supply-chain disclosures in its last 10-K filed before Liberation Day (April 2, 2025). Using the same model as our main coding pipeline (GPT-4o), with prompt provided in Appendix A6, each firm is scored from 0 to 1 on how much its production relies on manufacturing, assembly, components, or raw materials located outside the United States. The measure captures supply-chain structure, not the firm's expressed concern about tariffs.

As Table B12 shows, the upstream-hardware interaction is stable across specifications, ranging from -0.081 to -0.099 and significant at the 1% level throughout. None of the firm-characteristic interactions is statistically significant. The hardware coefficient declines modestly once import dependence is included but remains large and significant, consistent with import exposure being one component of mobilizability for hardware firms. In this sense, the divergence among upstream and downstream hardware firms is therefore not driven by differences in size, capital intensity, or their import dependence.

Table B12: Robustness to Pre-Treatment Firm Characteristics

	(1) Baseline	(2) + Size/Capex	(3) + Import Dep.	(4) + All Controls
Hardware × Post	0.184***	0.198***	0.148***	0.153***
	(0.042)	(0.043)	(0.053)	(0.057)
Upstream × Hardware × Post	−0.082***	−0.091***	−0.081***	−0.099***
	(0.025)	(0.032)	(0.023)	(0.031)
Log revenue × Post		0.009		−0.009
		(0.052)		(0.052)
Log assets × Post		−0.030		−0.023
		(0.045)		(0.047)
Capex intensity × Post		0.011		0.011
		(0.021)		(0.023)
Import dependence × Post			0.028	0.038
			(0.028)	(0.030)
Firm FEs	Yes	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes	Yes
Observations	486	486	486	486
Firms	65	65	65	65
Within R^2	0.086	0.090	0.091	0.098

Notes: Standard errors clustered by firm in parentheses. All firm covariates are pre-treatment, standardized, and interacted with the post-Liberation-Day indicator; firm fixed effects absorb their levels. Revenue, assets, and capital expenditure are for fiscal year 2024. Capital-expenditure intensity is set to zero for two asset-light firms (Airbnb, Workday) that do not separately report capital expenditures; results are identical if these firms are instead dropped. Import dependence (0–1) is coded from the supply-chain disclosures in each firm's last 10-K filed before April 2, 2025.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

To further examine the alternative explanation that import dependence drives firms' tariff stances, we interact pre-treatment import dependence with the post-treatment indicator, and add a three-way interaction between import dependence, upstreamness, and the post-treatment indicator (Table B13). Firms that are more import-dependent before Liberation Day are more negative about tariffs afterward. However, import dependence does not moderate the effect of upstreamness in the pooled sample. Among hardware firms, the point estimate suggests that the effect of upstreamness strengthens with import dependence, consistent with our product-level binning results, but this interaction is only marginally significant. Together with the evidence above, we have shown that the cost-structure explanation is valid but insufficient: import dependence accounts for part of the variation in tariff negativity, but not for why upstream hardware firms responded less negatively. This gives a strong merit to our geoeconomic explanation.

Table B13: Import Dependence and the Effect of Upstreamness

	(1) All Firms	(2) Hardware Firms
Upstream × Post	−0.018	−0.073***
	(0.026)	(0.019)
Import dependence × Post	0.066***	0.060***
	(0.023)	(0.018)
Import dependence × Upstream × Post	−0.007	−0.033*
	(0.028)	(0.017)
Firm FEs	Yes	Yes
Quarter FEs	Yes	Yes
Observations	486	243
Firms	65	32
Within R^2	0.051	0.075

Notes: Dependent variable is tariff negativity. Standard errors clustered by firm in parentheses. Upstreamness is standardized over the full sample. Import dependence (0–1), coded from the supply-chain disclosures in each firm's last 10-K filed before April 2, 2025, is standardized within each estimation sample, so the Upstream × Post coefficient is the effect of upstreamness at that sample's mean import dependence. Firm fixed effects absorb the levels of both variables and their product.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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